Oslo Metropolitan University

TREK 2.0 :
Bilateral Gait Monitoring System

image generated by AI

Completed by:
Chethan James Magnan,
Jeyapiriyan Thevaranjan,
Thomas Druillole,
Luke ten Klooster,
Michael Prediger,

Fall 2025

Mentored by:	Supervised by:
Tom Muir	*Dr. Ali Muhrtagolu*
Tengel Aas Sandtrø	*Dr. Haroon Khan*

European Project Semester

Table of Contents

1. Introduction

1.1. Project context

As of 2010, there were 1.6 million people suffering from hip fractures, with 600,000 in Europe alone [1]. This number is expected to rise, with researchers predicting 6 million hip fractures annually by 2050 [2]. For the average person, the process seems straightforward: they go to the hospital, the surgery is successful, and all is well. However, about 40 to 60% of patients never fully regain their previous activity levels, experiencing declining mobility in daily life [2]. The cause is often musculoskeletal impairment following a hip fracture (sarcopenia) and an asymmetric gait, where steps do not align during walking [3]. Without frequent monitoring, this condition tends to progress, leading to slower walking, more frequent falls, and overall deterioration in quality of life. Because this decline occurs gradually and recovery feedback is delayed, many patients do not adhere to physical therapy or notice improvements. Improved monitoring and timely feedback from healthcare professionals could help these patients recover daily function more effectively [4].

1.2. Project objectives and relevance

For patients recovering from musculoskeletal impairments, measuring gait asymmetry is essential to track recovery and prevent physical deterioration before it becomes severe. The device must be easy to use, non-intrusive in daily life, and provide actionable information. Some solutions exist on the market but do not fully address our use case: supporting older adults in a physical therapy context to recover daily function. Some devices, like the MbientLab MMRL [5], are affordable but provide only raw sensor data, which can overwhelm non-technical users. Others, such as the Movella Xsens Awinda [6], are designed for highly specific clinical research applications, relying on a distributed sensor grid within a controlled environment and are prohibitively expensive, with purchase details handled directly via sales teams. Neither solution allows continuous, real-time gait analysis throughout the day.

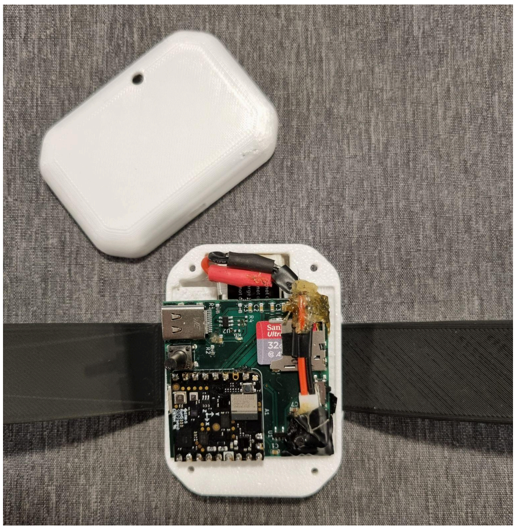

Figure 1.1 : Trek 2.0

Our project, Trek 2.0, is a compact wearable device designed to monitor walking patterns, building upon the prototype developed by the previous group. It enables patients and healthcare professionals to track gait asymmetry between the left and right legs in real time, providing insights to prevent long-term joint and muscle deterioration. The device is intended for use in physical therapy, offering continuous, easy-to-use, and non-intrusive gait monitoring to support recovery and improve daily function.

2. Scientific and Technical Background

This section starts by defining key concepts and providing an overview of the general research conducted in this area. This provides context, making it easier to understand the work of previous groups as well as our own development. More specific research and detailed studies can be found in their corresponding sections to maintain a clear and engaging flow.

2.1. Fundamental concepts and key definitions

General gait concepts :
Gait refers to the coordinated pattern of walking, describing how the legs and feet move during locomotion [7].

Gait assessment, also known as gait analysis, is the process of measuring and evaluating this walking pattern, typically using sensors such as inertial measurement units (IMUs) to quantify spatial, temporal, and symmetry-related parameters.

A gait cycle is defined as the interval between two consecutive heel strikes of the same foot, representing a complete sequence of movements that can be analyzed to extract relevant gait parameters. The different phases of the gait cycle are discussed in Section 6.3.2.

A gait event refers to a specific, identifiable moment within the gait cycle, such as a heel strike or toe-off. These events are key reference points for analyzing gait parameters, detecting asymmetries, and calculating stride and step characteristics.

Bilateral gait measurements refer to the simultaneous assessment of both legs, allowing for the evaluation of asymmetries and coordination between left and right limbs. These measurements are critical for understanding differences in gait patterns, detecting abnormalities, and improving the accuracy of gait analysis.

Stride length - definition and clinical relevance :

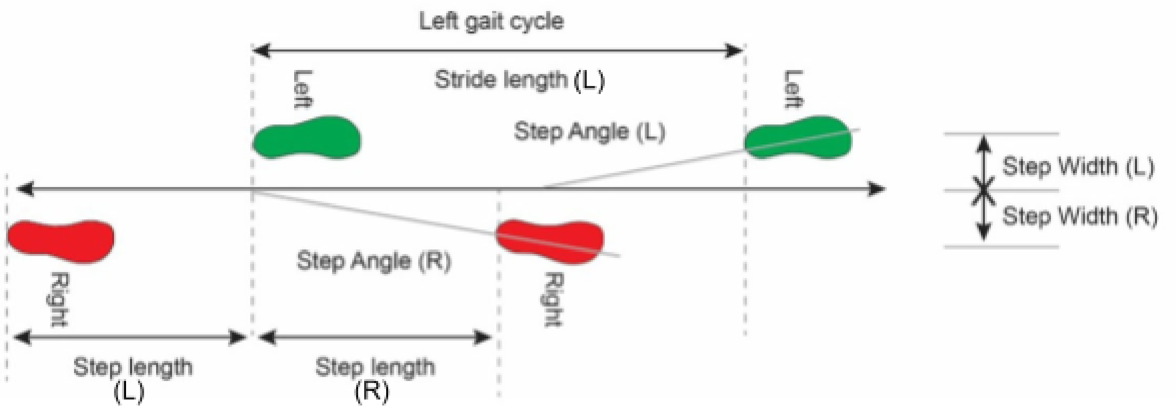

Figure 2.1 : Illustration of spatial parameters of gait

Stride length is a key gait parameter, defined as the distance covered by the foot during a single gait cycle. It is particularly relevant because it provides several advantages: it directly reflects walking efficiency and step consistency, offering a clear and interpretable measure of gait performance [8]. Furthermore, it can be tracked over time to assess changes or improvements in patients.

Stride length is sensitive to gait abnormalities and asymmetries, making it a valuable indicator for clinical assessment and rehabilitation. Studies conducted by Umeå University (Sweden), the University Clinical Centre of Kosovo, and St George's University (UK) have shown that stride length is strongly associated with clinical outcomes such as disability and mortality [8]. According to Wang and Bhatt (University of Illinois at Chicago), individuals with a history of falls exhibited shorter stride lengths and significant asymmetries, highlighting the potential of stride length monitoring in fall prevention [9].

Stride length can also be compared across individuals by normalizing it to patient height (normalized stride length = stride length (cm) / height (cm)). For instance, Bytyçi and Henein (2021) reported that this normalized stride length was more accurate in predicting adverse clinical events, with a summary sensitivity of 65%, specificity of 72%, and overall accuracy of 75.5% [8].

Other gait parameters, including stride time and cycle time, are presented in Section 6.3, where their calculation and use in deriving additional gait metrics are detailed.

2.2. Overview of existing methods and techniques

Several studies and projects have explored the use of inertial measurement units (IMUs) for gait analysis. These works provide insights relevant to the development of wearable gait monitoring systems.

IMU-based gait assessment and bilateral coordination:

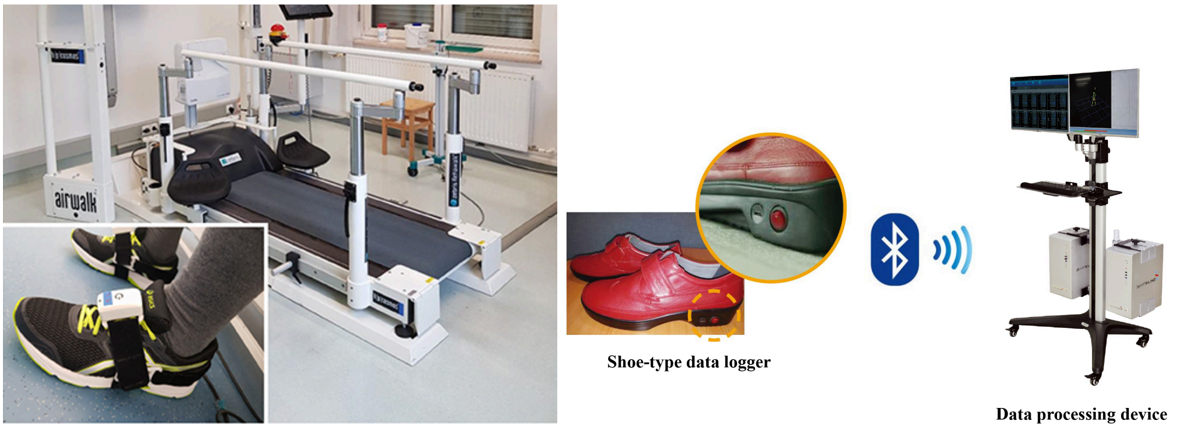

Figure 2.2 : Experimental setups from similar studies. Left: Laidig et al. (2021) – patient with foot-mounted IMUs and instrumented treadmill. Right: Han et al. (2019) – shoe-type IMU-based gait analysis system.

Laidig et al. (2021) [10] and Han et al. (2019) [11] demonstrated methods for assessing gait parameters using IMU sensors. Laidig et al. focused on foot-mounted sensors and calibration-free measurement of gait pattern, speed, rhythm, and step coordination, while Han et al. analyzed bilateral coordination and gait asymmetry with shoe-mounted sensors.

Sensor placement considerations:

Küderle et al. (2022) [12] and Panebianco et al. (2018) [13] investigated how sensor position affects measurement accuracy and reliability. Küderle et al. showed that different foot-mounted IMU placements strongly influence spatial parameter accuracy. Mean stride-length errors remained below 2 cm for most positions, while poorly located sensors, such as the heel, produced occasional single-stride errors up to ~60 cm, highlighting the importance of proper placement.

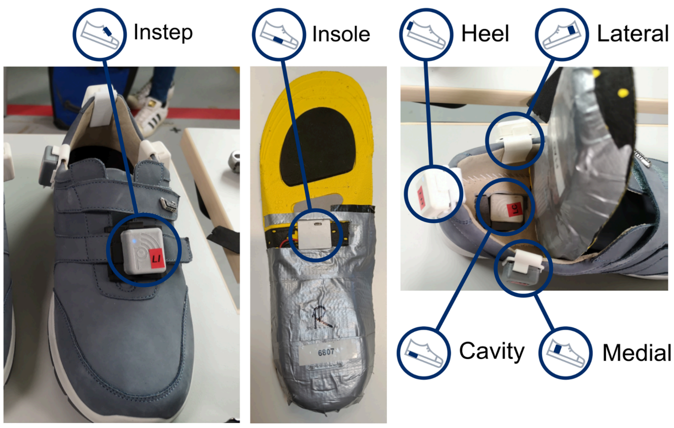

Figure 2.3 : Panebianco et al. (2018) – Sensor placement of the six shoe-mounted IMUs.

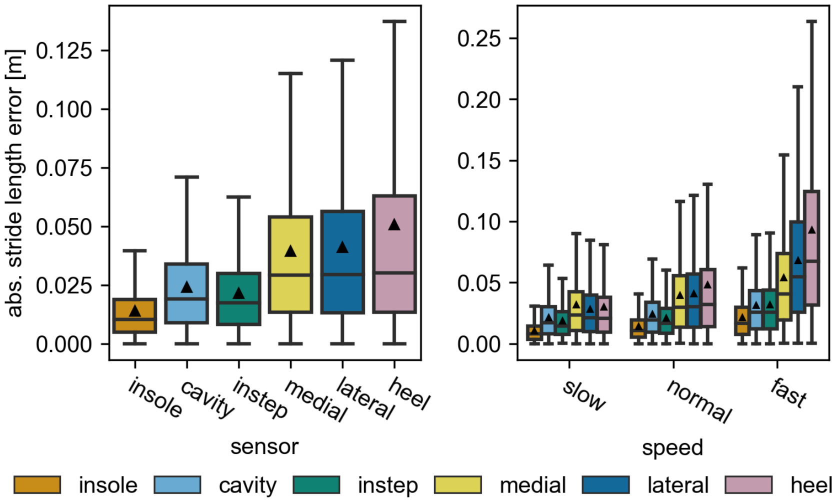

Figure 2.4 : Panebianco et al. (2018) – The absolute stride length error of each sensor position.

Panebianco et al. confirmed that sensor position significantly affects measurement repeatability, as shown in Figure 2.4, which illustrates the absolute stride-length error for each sensor position.

Gait event detection algorithms:
Panebianco et al. (2018) [13] evaluated 17 different algorithms for detecting gait events and estimating temporal gait parameters from IMU data. Their results showed that shank- and foot-based algorithms were more accurate and repeatable than lower-trunk-based approaches for detecting heel strikes and toe-offs. Angular velocity-based methods outperformed acceleration-based ones for these events, providing more reliable measurements.

Gait symmetry and coordination insights:
Błażkiewicz et al. (2014) [14] and Gimmon et al. (2018) [15] focused on gait symmetry and coordination. Błażkiewicz et al. compared methods for quantifying left-right differences, while Gimmon et al. studied changes in bilateral coordination with age, using the phase coordination index and gait asymmetry metrics. These works provide guidance for selecting symmetry metrics and interpreting variations across different populations.

2.3. Relevance to our project

The reviewed studies provide practical guidance for the development of our wearable gait monitoring system. Although Laidig et al. [10] and Han et al. [11] focused on foot- and shoe-mounted sensors, resulting in data different from ours, which uses lower-limb bracelets, these studies can serve as benchmarks to help interpret sensor signals and guide the measurement of gait parameters. They also provide a useful comparison for evaluating our final gait measurements.

Küderle et al. [12] and Panebianco et al. [13] emphasize the critical role of sensor placement and algorithm selection in ensuring measurement accuracy and repeatability. Insights from these studies support our choice of bracelet position. They can also guide algorithmic decisions for detecting gait events such as heel strikes and toe-offs.

Błażkiewicz et al. [14] and Gimmon et al. [15] are relevant to the project because it involves calculating differences between the left and right foot during walking. Błażkiewicz et al. provide guidance on selecting appropriate gait symmetry coefficients to quantify these differences, while Gimmon et al. show how bilateral coordination and gait symmetry vary with age. Understanding these variations helps interpret IMU sensor data accurately and assess gait differences across different populations.

3. Review of Previous Work

3.1. Previous Trek device

This project is a continuation of the previous Trek device, which presents both advantages and challenges for development.

Figure 3.1 : Final prototype of the previous Trek device [16]

The previous Trek device is a proto-board with an IMU, (Inertial Measurement Unit), BLE transmitter and receiver (Bluetooth Low Energy), MicroSD Card, charging port, and power switch. This device takes accelerometer, gyroscope, and magnetometer data and saves this to a MicroSD card. The data can then be uploaded offline into an accompanying MATLAB script to calculate and generate a plot for the asymmetries between each leg.

Due to the existence of the previous product, we should be able to achieve more and better address the clients' needs. We were starting with the basic circuitry, programming, mobile communication, and classification algorithms from the previous group. This would mean we could focus on optimization and expansion.

3.2. Strengths of previous device

The previous Trek device is the first step forward and accomplished what they set out to do - the device has a completed data path, the overall device physically worked. It is an end to end device, able to receive input from the Nicla, send these over the soldered wires for storage in a MicroSD card, and then output a useful gait asymmetry calculation with graph.

The main functional units were logically decided and purchased with off the shelf components to make the prototype work as predicted for the final use case. Finally, the previous team did use Github sharing how they had structured the arduino measurements and left a report detailing their project planning, execution, and suggestions for the future. This report was particularly useful as it allowed us to understand how they thought through their device, recreate the overall circuitry as photos covered how the device was wired, and have a very clear goal.

3.3. Limitations

There are three main areas for improvement based on the deliverables we received from the previous Trek group :

- Real-time asymmetry calibration was not possible. The device stored data on the SD card, and asymmetry values were computed offline in MATLAB. This prevented immediate feedback for the user.
- The device did not include any user interface. As a result, patients could not adjust their walking based on real-time feedback, and sharing data with a clinician was inconvenient. Without an interface, a medical professional could not easily compare changes in daily mobility, identify whether physical decline had slowed, or evaluate how gait evolved throughout the day.
- The device was too large. A more compact design was requested to improve usability and reduce manufacturing cost. Smaller devices are also better accepted by older adults. This is supported by recent long-term wearable research, which reports that "almost without exception, participants desired smaller and lighter wearables or expressed discontent about the size and weight of the device they were asked to wear in the study" [17].

3.4. Improvement and development strategy

The initial Trek device aims to solve this with a Goldilocks solution. The device is portable and customized to detect gait, but as we address in depth in the previous section, it is not real time, compact, nor have a ui device and therefore still in a prototype phase. Our further developments to the Trek device aims to make it a professional gait-tracking tool. We want it to be used for assisting in physical therapy, where professionals may monitor patient walking patterns in real time to prevent long-term joint and muscle damage.

With this objective, we defined three main development directions: creating a mobile interface, enabling real-time IMU data transmission, and reducing the device size and weight.

Achieving real-time communication requires redesigning the wireless data flow and reworking the embedded signal processing. To reduce energy consumption and extend battery life (thus allowing a smaller battery), most calculations must be performed directly on the embedded system, whereas the previous group relied on heavy MATLAB-based offline processing, which is not feasible on an embedded system.

We also plan to develop a custom circuit board, implement improved data batching to limit power draw, and design a new, thinner enclosure, so that we can reduce the overall footprint of the device and its size. This new design will also provide additional benefits in terms of manufacturing cost and component integration.

Figure 3.2 : Comparison image between TREK 2.0 (on the top) and TREK 1.0 (below)

4. Project Management and Task Organization

4.1. Mission Goals Scope (MGS)

Mission:
Trek is a clinical gait-tracking tool for physical therapy where professionals monitor patient walking patterns in real time to prevent long-term joint and muscle damage. Our mission is to enhance the system by improving user comfort, optimizing both hardware and software performance, and designing an intuitive interface that allows patients to monitor their gait in real time.

Goals:
Our main focus is completing the client's request; make Trek real time, more compact, and have an improved user interface. For a functional TREK system capable of bilateral gait monitoring in real time we need system to turn on and send basic imu measurements via wireless communication

To make it lighter and compact (at least 20% smaller and weigh 20% less), we should have a custom circuit board design that is 20% smaller than previous as well as a case that's 20% smaller by volume.

Finally, to improve the value of the user experience, we should make a mobile user interface that can assess and process the system in real time. This includes an adjustable window time scale to see a whole day of measurements.

Scopes:
Our main constraints are the previous group's work and the lack of clinical use case. We will not remake the project from the beginning and should reuse as much as possible from the previous group's work, which means some of their decisions or technical debt must be considered.

This device is also not for a clinical use case at the moment. The intention is not for the product to be commercialized, nor diagnose medical problems, nor implement a user account management system. A commercialized product would need far more polishing and testing with user groups. To diagnose medical problems, we would want to work closer with medical professionals and need far more time and funding. Finally, as the group lacks a data specialist, the development of a digital database for the online interface will be omitted.

4.2. Project organization and planning

For the management part of our project, we wanted to ensure that the work is well organized and that every team member knows their responsibilities. Management therefore plays a key role in our team. During the research phase, we defined in detail the tasks required to complete the project. We grouped these tasks into five main study domains, each with clear objectives :

- Redesign circuit board: improve compactness, update components, and optimize embedded functionality.
- Redesign case: enhance ergonomics, adapt to the new materials and reduce size and weight.
- Sensor data processing: implement efficient algorithms for gait analysis in real-time.
- Wireless communication: enable real-time BLE data transfer between device and UI.
- User interface: develop a mobile interface for easy monitoring and feedback.

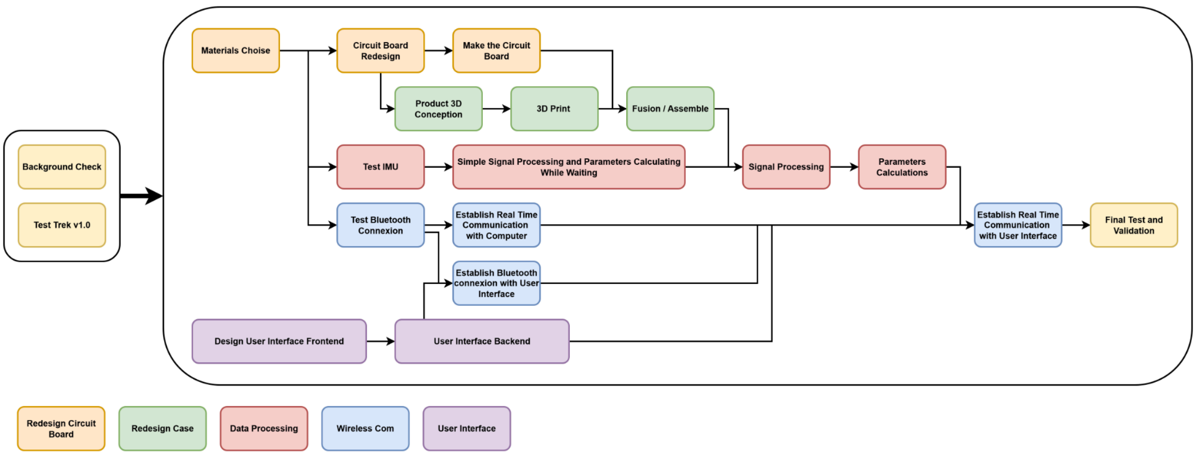

Figure 4.1 : simplification of the GANTT diagram

To manage the timeline, we created a detailed Gantt chart that helps us plan the various tasks, track progress, and identify deadlines (see appendix). We also prepared a simplified version of the Gantt chart (Figure 4.1) that groups the most important stages of our project. This version makes it easy to see the dependencies between stages.

To ensure the smooth progress of the project and clearly define each team member’s responsibilities, our team was organized around complementary areas of expertise. Each member took charge of specific aspects of the development, from mechanical design to software and embedded systems engineering, ensuring that all parts of the project were properly coordinated and progressed according to schedule.

- Chethan James Magnan – Computer Engineer: responsible for the circuit board redesign and communication with the supervisor.
- Jeyapiriyan Thevaranjan – Embedded Systems Engineer: focuses on embedded firmware development and manages the Gantt chart to keep the project on schedule.
- Thomas Druillole – Mechanical Engineer: handles Bluetooth real-time communication and backend support for the user interface.

- Luke ten Klooster – Mechanical Engineer: responsible for frontend and user experience.
- Michael Prediger – Mechanical Engineer: manages CAD design and prototyping of the case.

4.3. Communication and collaboration

To ensure effective communication and collaboration, our team relies on Microsoft Teams, WhatsApp, and GitHub. These tools allow us to share ideas quickly, centralize contributions, and maintain easy access to all project materials. GitHub is particularly useful for tracking changes in code and documentation, while Teams and WhatsApp facilitate fast discussions and coordination.

Although each team member has specific responsibilities as a team leader in their area, we work closely together rather than independently. Team leaders coordinate tasks, but all members are involved whenever needed, ensuring that progress is smooth and consistent across all areas.

Weekly meetings are held to review the work completed, discuss challenges, and plan tasks for the coming week. At the beginning of each meeting, members give a brief update on their progress. All discussions and task assignments are recorded in a shared document, which allows everyone to stay informed, follow the project's progress, and adjust priorities if necessary.

This structured approach to communication and collaboration helps our team stay aligned, efficiently distribute the workload, and maintain high productivity throughout the project.

5. System Architecture

5.1. Hardware design and architecture

The systems hardware architecture organizes how power (via the battery and its USB C charging circuit) will be supplied to the functional units of the Trek device (the Arduino Nicla with its BLE and the MicroSD card) as well as how the user will interact with the device (the BLE on off button and the power switch). This overall architecture is illustrated in the figure below. By optimizing the design for production, the cost is further reduced from approximately 130 EUR to 70 EUR. While the Arduino cost remains around 60 EUR, the price of the peripherals drops to roughly 15 EUR when manufactured in quantities of 1000.

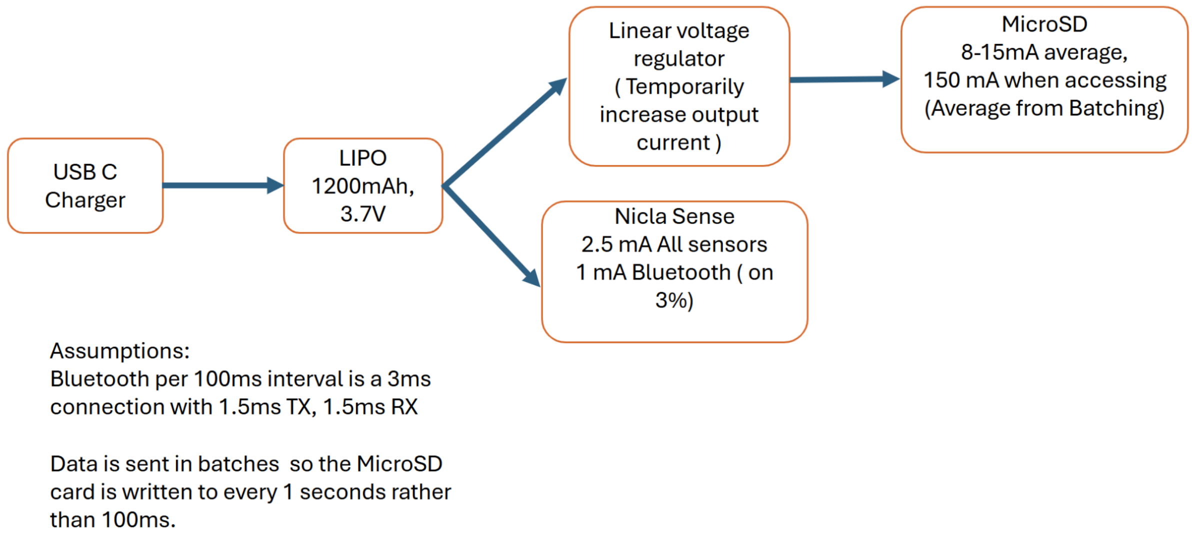

Figure 5.1 : Logical schema of the hardware architecture

The previous group used many of the same type of component and the overall structure is fairly similar. This was maintained to use the previous groups contributions as we agreed with their rationale - this component setup offers the main functionality such a device would require and met our clients requests.

The only major modifications to the circuit were; the BLE button, decreasing the distance between each component, and updating components that went out of stock. Decreasing the component distance makes the entire device more compact and leads to less noise. We achieved this through the design and outsourced manufacture of the Printed Circuit Board (PCB). Finally, some components were no longer offered by the suppliers and as such were updated with equivalent but more recent versions.

We also disagreed with the previous group's battery calculations and performed our own.By slightly reducing power consumption—primarily by decreasing the access interval to the MicroSD card (configured in the embedded firmware), we calculate that we can power the entire board per day on a minimum 3.7 V, 83 mAh, 307 mWH Lithium polymer (LiPo) battery.

For our design, we chose to use the 450 mAh 3.7v Utuav battery due to its thinner horizontal form and multiple day charge. This is again beneficial as it continues to make the design more compact for our end user when compared to previous 1200mAh LiPo. In terms of volume, the battery footprint decreases from 10.8 cm³ to 7.14 cm³ (a 34% reduction), and the surface area required inside the device decreases from 21.6 cm² to 10.2 cm² (a 53% reduction).

5.2. System logical architecture

Our TREK 2.0 system is built on a technical architecture that can be divided into three levels:

1) Data acquisition and real-time processing on the embedded board (Arduino Nicla Sense ME):

The embedded firmware continuously collects IMU data. These signals are processed with filtering methods to identify gait phases and spatial positions. Only the essential information—such as stride length and asymmetry—is extracted in order to reduce transmission load and optimize energy consumption.

2) Low-latency wireless transmission via Bluetooth Low Energy (BLE):

The processed data are sent to the interface through BLE in small, low-latency packets. In parallel, all raw and processed data are stored on the MicroSD card to ensure full backup even in case of disconnection, allowing the system to operate without requiring a permanent Bluetooth connection.

3) Real-time visualization and parameter display through the user interface:

The interface receives and decodes BLE packets in real time, updates its graphs, and recalculates parameters over the time window selected by the user. This enables continuous and instantaneous visualization of gait behaviour and asymmetry between both legs. The user can choose between a minimal mode (real-time essential information: stride length, asymmetry) or a debug mode (detailed analysis, not real-time).

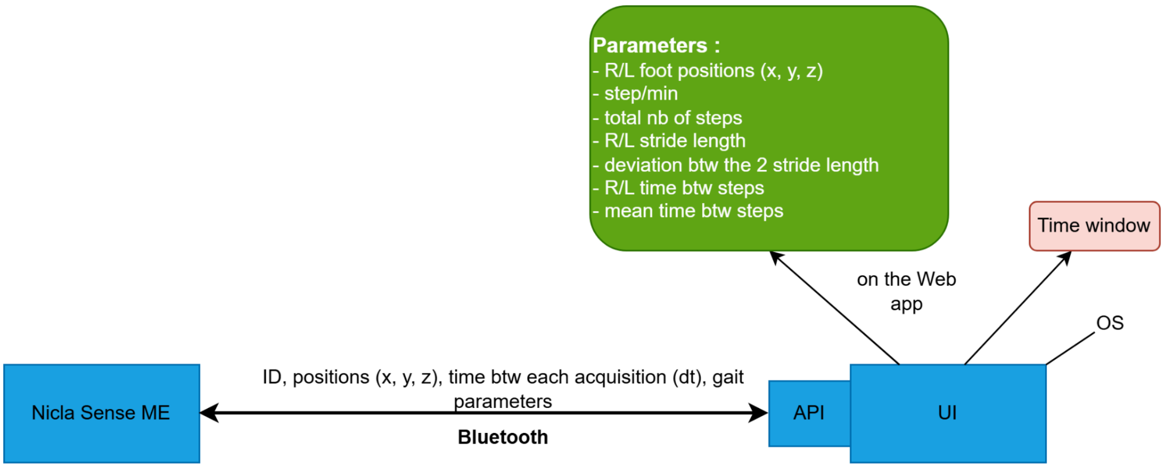

Figure 5.3 : Logical schema of our device architecture

5.2. Embedded firmware

The Nicla Sense ME embedded in the bracelet must operate fully autonomously to ensure continuous gait monitoring under all conditions. For this reason, the system requires a robust embedded firmware capable of handling all operational scenarios without external intervention.

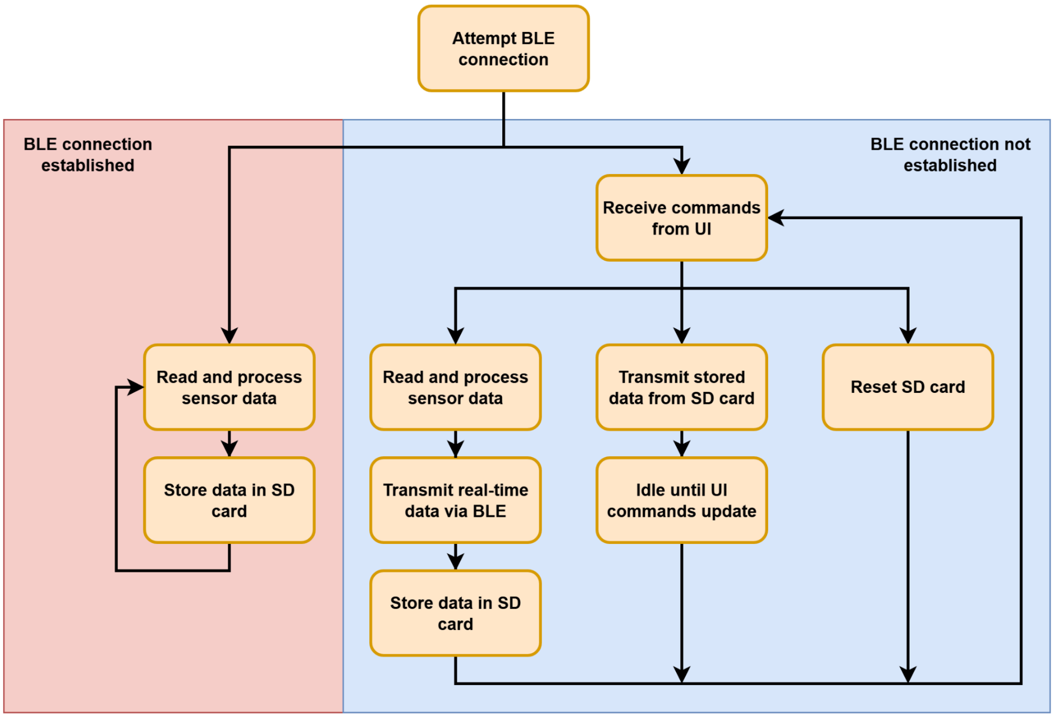

Figure 5.2 : embedded firmware architecture overview

Real-time communication is one of the major improvements compared to the previous group's work. At startup, the firmware attempts to establish a Bluetooth connection with the user interface (UI) during a 20-second initialization window. Two operational modes are then possible:

- If the Bluetooth connection is not established within the allotted time : the system switches to standalone mode, where gait measurements are continuously acquired and stored locally on the MicroSD card.

- If the Bluetooth connection is successfully established : the firmware enters real-time streaming mode. Sensor data are transmitted live to the UI while simultaneously being logged on the MicroSD card to ensure that all data are saved.

  In this mode, the board can also receive configurations from the UI, such as the selected operating ui-mode (section 6.5) and the data transmission frequency (section 6.5), allowing the user to adapt system behavior in real time.

  Additionally, when the device receives the command to enter debug mode and the onboard button is pressed, all previously stored MicroSD-card data are transferred to the UI.To return to streaming mode, the ui-mode must be switched and the button pressed again.

This dual-mode architecture guarantees continuous gait data acquisition regardless of connectivity conditions.

6. Implementation

6.1. PCB design and hardware design

For the Printed Circuit Board (PCB) design and function, we used the existing circuit. Time constraints and Oslo Met's electronics lab dictated the scale and type of PCB. We did surface-mount full assembly with Eurocircuits, so the main focus was designing the PCBs and validating or modifying them based on tests.

The design process proceeded relatively smoothly, taking a few weeks to find the original parts used by the previous group. We chose to maintain the same PCB and general components to prevent any coding or peripheral communication errors. As most of our initial development was done using the Arduino Nicla given by the previous group, we kept using this as our main CPU / processing unit.

The PCB functions as a socket connecting the Arduino Nicla sense me and attaches the battery charger, battery pack, MicroSD card with reader, and two switches – a push button for choosing to access Bluetooth as a power saving feature and an SPDT power switch. The designs were done in Kicad, and some sampling from online schematics from Soldered Electronics was done (both are open source and available for our use case). The schematics, PCB layout, and 3D visualizations are included in the appendix.

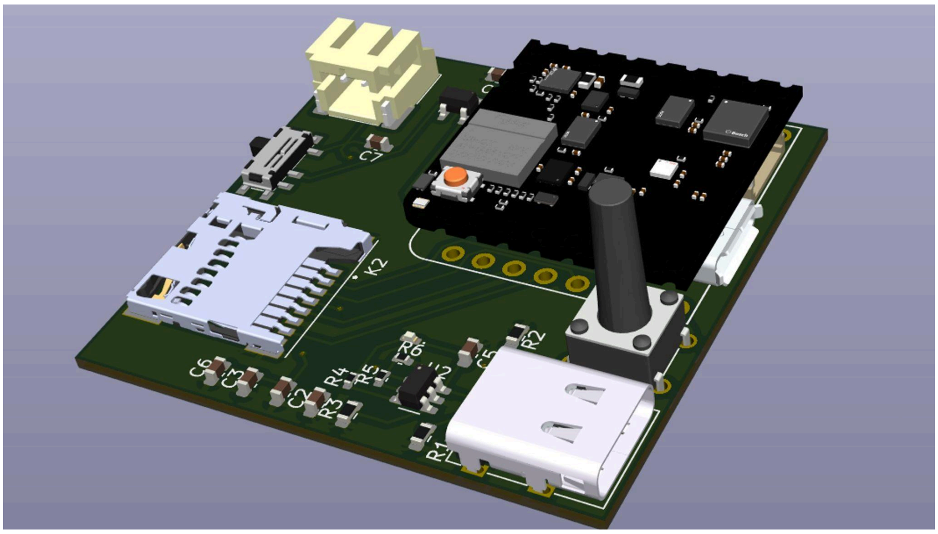

Figure 6.1 : CAD of the PCB board

These designs were then reviewed and submitted to Eurocircuits for manufacture. After receiving 5 PCBs, a multimeter and power supply were used to confirm that there were no shorts in the circuit, that each functional unit worked appropriately, and that the PCB could receive and charge power from the lithium ion battery.

Some minor modifications were necessary: the button was rotated, the power connector needed to be modified to accommodate our battery's connector, and a resistor needed to be removed that shorted the linear voltage converter for the USB-C charger. Once these modifications were complete, the PCB met the needs of the project in terms of battery life, support for real-time functionality, and size reduction.

The overall battery life matched the theoretical calculations. Real-time access to the MicroSD card is supported, and Bluetooth communication remains fully functional. Finally, the PCB footprint was reduced from 21.46 cm² to 17.22 cm², corresponding to a 19.8% reduction.

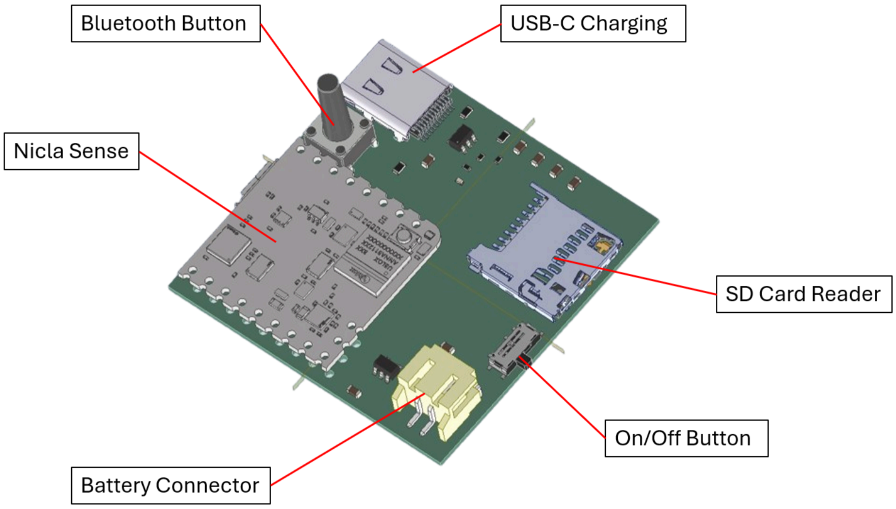

Figure 6.2 : CAD of the PCB with component labels

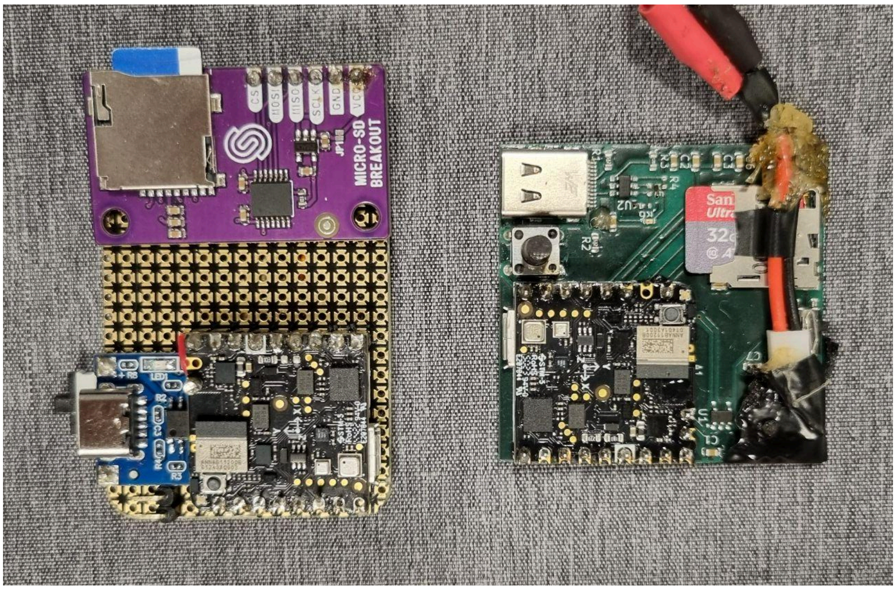

Figure 6.3 : Previous board (left) compared to our PCB (right)

6.2. Mechanical and 3D design

This part of the report outlines the development process of the 3D model and the manufacturing of the prototypes. The focus of this work was the design of a new case that securely positions all internal components while improving overall usability. Compared to the device created by the previous group, our prototype aims to offer enhanced wearing comfort. The goal was to create a compact and lightweight design that provides mechanical stability.

6.2.1. Manufacturing methods

Before selecting the materials, suitable manufacturing methods must be determined.

Conventional Manufacturing :

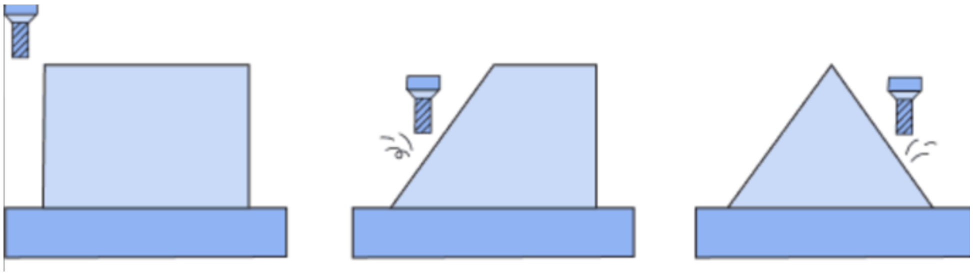

Figure 6.4 :Conventional manufacturing schematically

Conventional manufacturing processes are traditional methods that primarily involve machining, forming, or separating techniques in which material is mechanically processed, such as turning, milling, drilling, or grinding. As shown in the figure, you can see what a milling process looks like: you start with a solid block of material and then remove material until the final part is created.
For manufacturing our prototype, this method is unsuitable, as the geometry is very challenging to produce with it.

Additive Manufacturing :

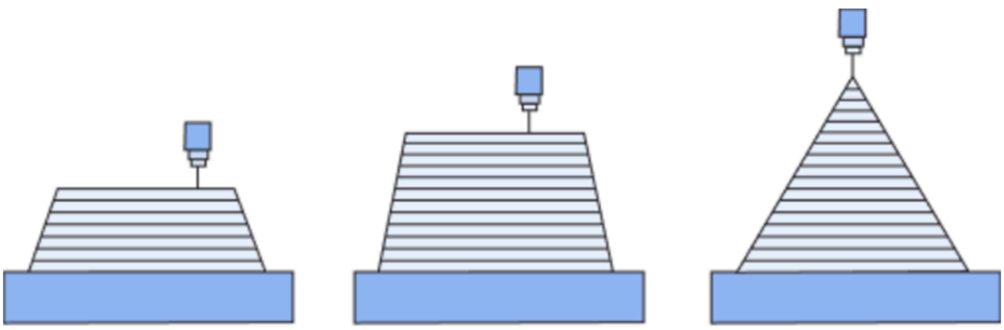

Figure 6.5: Additive manufacturing schematically

Additive manufacturing describes manufacturing processes in which components are built up layer by layer. Instead of removing material, as in conventional methods, material is added in a controlled manner, creating entirely new design possibilities. Figure 6.5 shows schematically how a component is built additively in layers.

Additive manufacturing is particularly suitable for prototyping, as it offers greater geometric freedom compared to conventional methods. Moreover, it allows components to be produced directly from CAD models without the need for additional tooling or extra costs. For this reason, we decided to use additive manufacturing processes, specifically filament-based 3D printing (FDM).

6.2.2. Material selection

After choosing the manufacturing method, we can choose the materials. Since our project is currently at the prototype stage, materials suitable for additive manufacturing are being considered. For series production, more cost-effective materials and processes such as injection moulding can be used to enable efficient manufacturing at larger volumes.

Case :
Thermoplastic polymers are particularly suitable, as they are lightweight, cost-efficient, and compatible with additive manufacturing methods such as Fused Deposition Modeling (FDM). Common materials include PLA (Polylactic Acid), ABS (Acrylonitrile Butadiene Styrene), and PETG (Polyethylene Terephthalate Glycol-modified).

PLA is easy to process and environmentally friendly but tends to be brittle and less heat-resistant. Whereas, ABS offers improved mechanical toughness and thermal stability but requires controlled printing conditions due to its tendency to warp. But we decided to use PETG which combines the advantages of both materials. It provides good impact resistance, higher chemical stability than PLA, and is relatively easy to print, making it a suitable material for functional prototypes.

The two tables below show the required printing settings and the mechanical properties of the filament types in comparison.

	PLA	ABS	PETG
Nozzle Temperature	190-220 °C	220-250 °C	220-250 °C
Bed Temperature	50-60 °C	90-110 °C	70-90 °C
Enclosure	Optional	Required	Optional
Cooling Fan	High (100%)	Minimal (0-20%)	Moderate (30-50%)
Bed Adhesion	Good adhesion to PEI or with glue stick; minimal warping	ABS slurry recommended; significant warping	Glue recommended due to excessive adhesion

Figure 6.6 : printing settings required for PLA, ABS and PETG [19]

	PLA	ABS	PETG
Young's Modulus	2641 ± 328 (MPa)	1685 ± 81 (MPa)	1986 ± 268 (MPa)
Bending Modulus	3279 ± 134 (MPa)	25.1 ± 0.5 (MPa)	44.3 ± 1.3 (MPa)
Tensile Strength	47.1 ± 1.2 (MPa)	1965 ± 72 (MPa)	1165 ± 61 (MPa)
Impact Strength	3.1 ± 0.3 (kJ/m²)	2.6 ± 0.4 (kJ/m²)	4.7 ± 0.4 (kJ/m²)
Elongation at Break	5.1 ± 0.3 (%)	8.2 ± 0.4 (%)	7.2 ± 1.1 (%)

Figure 6.7 : Comparison of material properties for PLA, ABS, and PETG [19]

Strap :
The strap used to secure the device around the ankle must combine flexibility, mechanical strength, skin compatibility. For the functional prototype in this project, the strap is produced using 3D printing with TPU (thermoplastic polyurethane), because the material combines suitability for elastic structures, advantageous mechanical properties, and cost-effectiveness.

6.2.2. 3D Design

The mechanical components of the prototype were designed using Autodesk Inventor. The following figures illustrate the final 3D design.

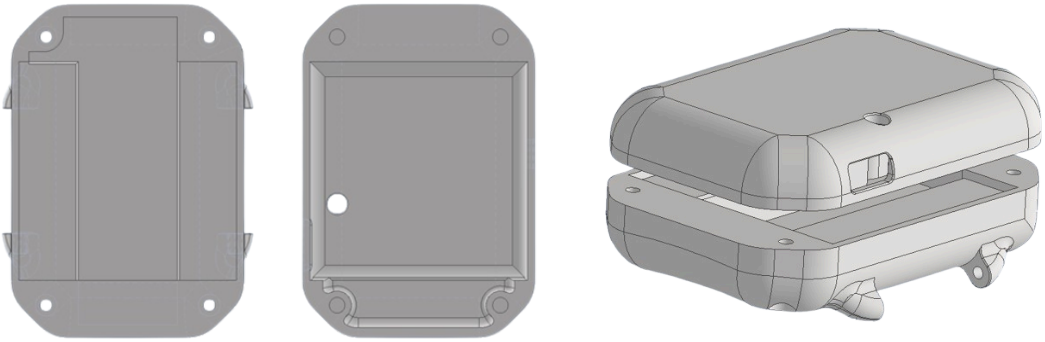

Figure 6.8 : left: Top view top and button case parts, right: Opened view of the case parts

In the final version of the prototype, the battery is clamped into the bottom of the case. The electronics are placed on the battery and held in position by the top part of the case. The top part features a hole for the Bluetooth button, while one side has a cutout for the On/Off button and the opposite side provides an opening for the USB-C charging port. The two parts of the case are fastened together at all four corners with screws.

Figure 6.9 : top view of bottom case part

The underside of the case that rests against the leg is slightly curved to better fit the shape of the leg. All four corners of the case are chamfered, and all external edges are rounded. This design not only makes the component more comfortable to wear and handle, but also improves its stability, as chamfers and radii help distribute forces more evenly.

The case also includes attachments for the strap at the bottom. The strap is designed similarly to a watch strap. One side features a pin buckle, while the other side has the holes for fastening.

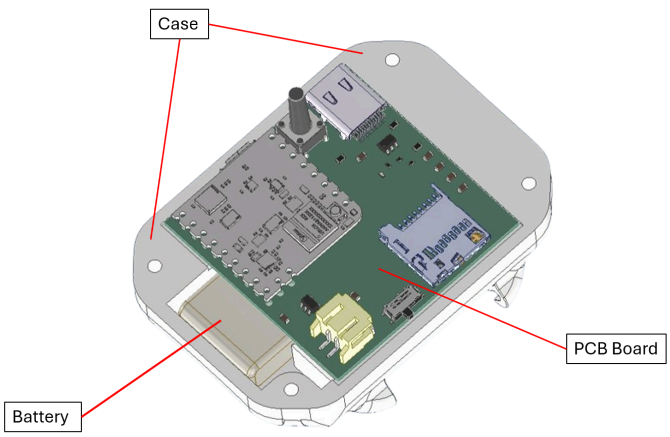

Figure 6.10 : Assembled CAD model of case, battery and PCB board

The developed prototype successfully improves upon the device created by the previous project group in terms of size, weight, and cost efficiency. The redesigned case reduces the overall volume from 933,8 cm^2 to 718 cm^2, corresponding to a decrease of approximately 23,12%. This more compact geometry not only enhances wearing comfort but also minimizes interference with natural ankle movement.

Similarly, the total weight was lowered from 105,8 g to 75,1 g, representing a reduction of around 29%. This is particularly relevant for continuous gait analysis, where every additional gram affects fatigue and user acceptance.

Overall, these improvements make the device significantly more comfortable, lightweight, and user-friendly, contributing to better long-term wearability and a more reliable data collection experience.

6.3. Signal processing and algorithm development

6.3.1. Data acquisition

Similar to the previous group, data collection is performed using the Nicla Sense ME board through comparable methods. This board integrates the BHI260AP [20], a 6-axis IMU (3-axis accelerometer and 3-axis gyroscope). The sensor operates at 1.8 V with a standby current of 8 µA, resulting in a very low power consumption of 0.014 mW in standby mode [21]. In this mode, the sensor remains active and capable of basic measurements while consuming minimal power, enabling the board to operate autonomously without a constant connection to a computer. The device also includes a smart AI sensor capable of on-board motion detection and sensor data fusion, which is discussed in a later section.

Different types of data can be collected from the Nicla Sense ME board [22]. For the gyroscope, SENSOR_ID_GYRO is used because it provides calibrated measurements with offsets corrected, ensuring accurate angular velocity readings. Similarly, accelerometer data could be retrieved from the corrected accelerometer channel SENSOR_ID_ACC, but this still includes the effect of gravity. For this reason, SENSOR_ID_LACC is preferred, as it delivers linear acceleration with the gravitational component removed and internal filtering applied, providing cleaner signals that better represent actual motion.

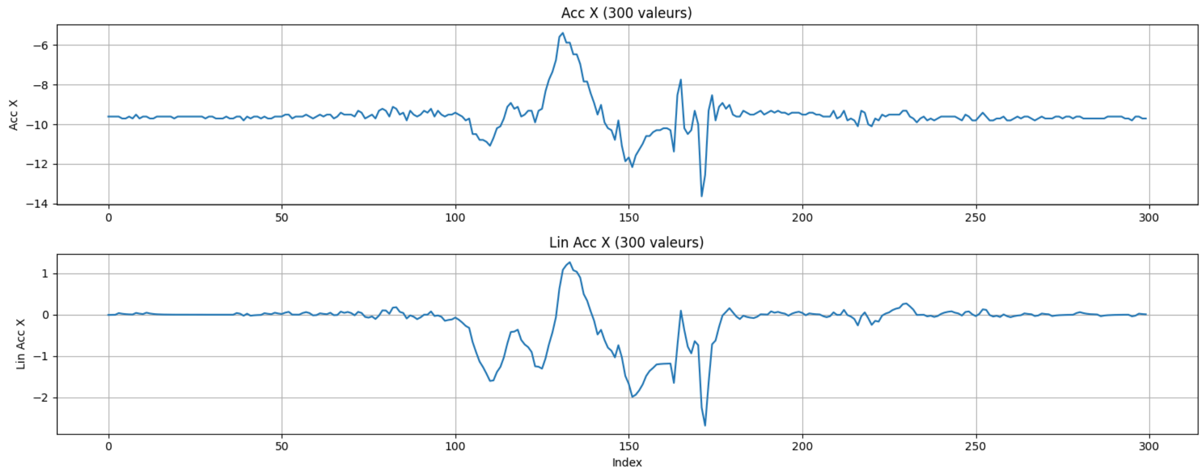

Figure 6.11 : comparison of raw and linear x-axis acceleration showing gravity removal and low-frequency noise reduction

6.3.2. Gait State Analysis

Detection of immobility :
It is important to detect immobility at two levels. First, identifying when the user is completely stationary helps avoid unnecessary calculations. Second, detecting when the foot remains still between steps allows variables such as velocity and position estimates to be reset for each step. This prevents error accumulation over time and improves the accuracy of step-by-step motion analysis. The following figure illustrates how such resets can significantly improve data quality.

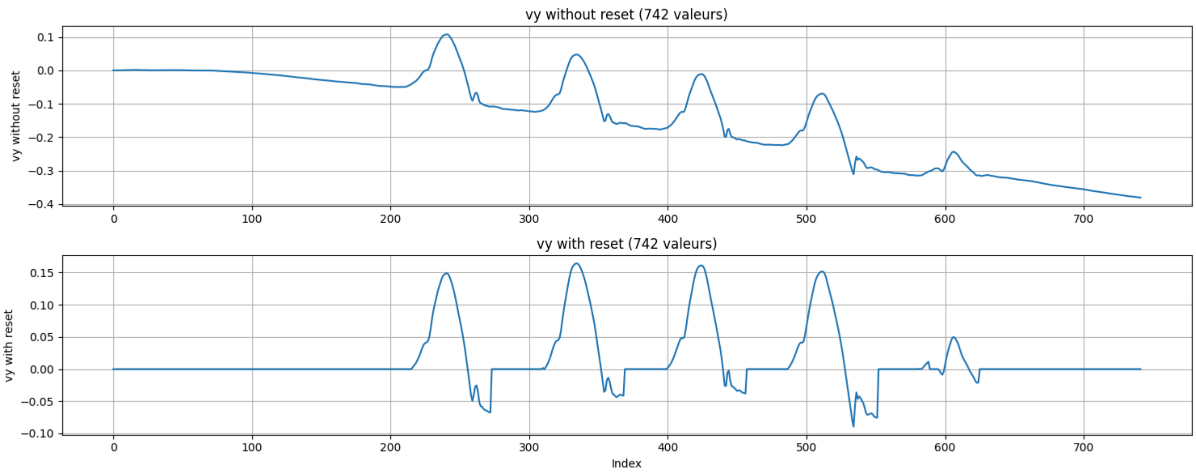

Figure 6.12: Effect of step-by-step reset on velocity measurements

To detect foot immobility, the previous group used a simple calculation based on the magnitude of the three-axis accelerations :

$$\mathrm{Mag} = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Since they worked with raw accelerometer values (SENSOR_ID_ACC), the magnitude remains around 1 g (accMagnitudeNorm) when stationary, reflecting the constant gravitational acceleration. Immobility was thus detected using the condition :

fabs(accMagnitude - accMagnitudeNorm) < accThresholdTolerance

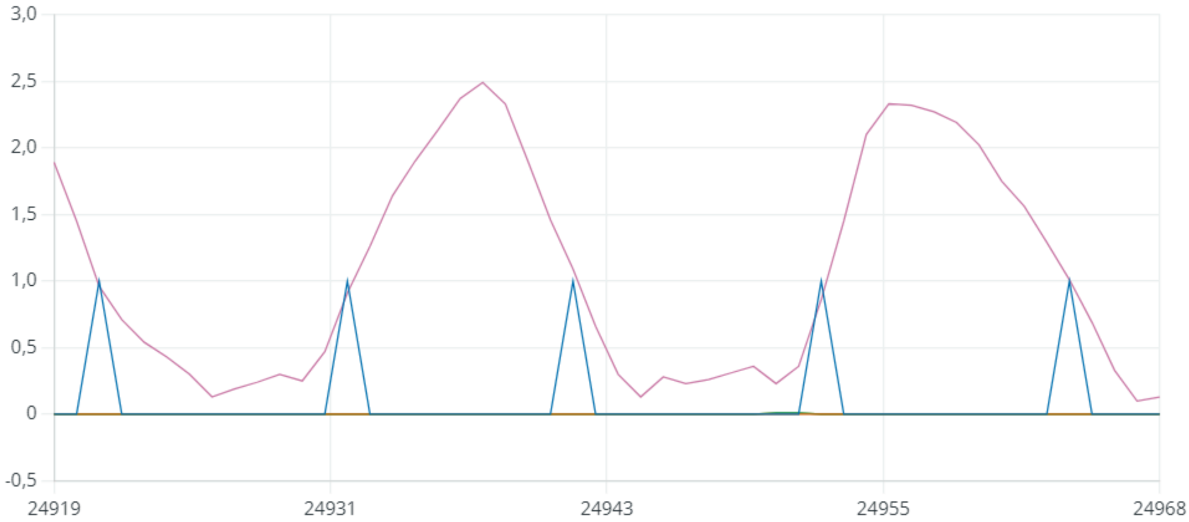

Figure 6.13 : Foot immobility detection from the previous group – acceleration magnitude (pink) and reset activation (blue)

However, this approach has several limitations. As shown in the figure, even when the board is in motion, the acceleration magnitude can easily pass through the immobility zone near 1 g. This can cause the reset activation to occur during walking, potentially introducing errors in velocity and position calculations.

To improve immobility detection, the board is considered stationary only when the acceleration magnitude remains within the immobility zone for a prolonged period. Instead of using the magnitude directly, its derivative is computed:

```
float deriv = (accMagnitude - prev_acc) / DeltaTime;
```

A smoothed derivative is then obtained using an exponential moving average (EMA) over the last two derivatives:

```
float ema_deriv = alpha_deriv * deriv + (1 - alpha_deriv) * prev_ema_deriv;
```

The EMA provides a weighted average of past values, with weights decreasing exponentially for older data. The parameter alpha_deriv controls the responsiveness:

small alpha → strong smoothing, past values dominate
large alpha → weak smoothing, recent changes dominate

An alpha of 0.3 was chosen, giving 70% weight to the previous EMA, so the smoothed derivative reacts slowly to rapid variations. The new derivative contributes only 30%, reducing instantaneous peaks and noise. This approach allows ignoring rapid fluctuations caused by accelerometer noise while still detecting actual movement. A moderate alpha (0.2–0.3) balances responsiveness and smoothing.

In addition to the derivative, the acceleration magnitude was also smoothed using an alpha of 0.2, allowing a dual condition for immobility detection:

```
bool near_ref = fabs(ema_acc - acc_ref) < tol_acc;
bool slow = fabs(ema_deriv) < deriv_tol;
if (near_ref && slow && !was_still) {
        //reset
}
```

Thus, the foot is considered immobile when the magnitude is close to 1g and the derivative is near zero. However, since linear acceleration is used directly, the reference magnitude is set to 0. During movement, the magnitude rarely approaches the immobility zone, but to ensure robustness, the derivative check is retained as an additional verification.

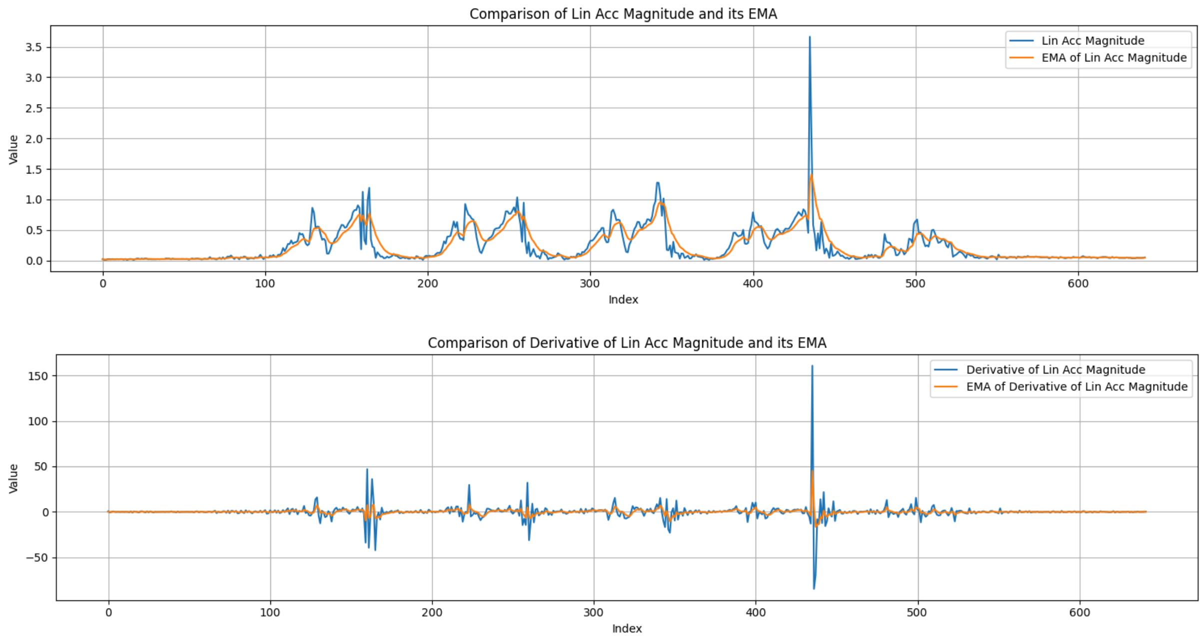

Figure 6.14: Comparison of linear acceleration magnitude and its derivative with their smoothed EMAs

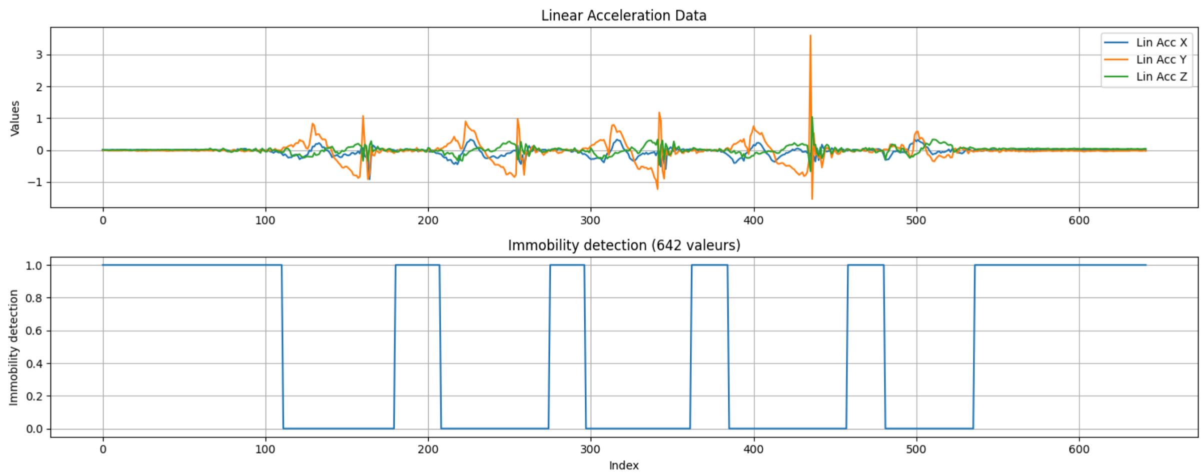

Figure 6.15 : Linear x-acceleration and foot immobility detection

Figure 6.14 illustrates the effect of EMA smoothing and shows how acquisition errors and abrupt fluctuations can be effectively removed. Figure 6.15 demonstrates the performance of the immobility detection algorithm: the signal switches to true (1) when the board is stationary and to false (0) when it is in motion.

Gait state identification during walking :
To calculate accurate gait measurements and estimate positions, it is necessary to understand the structure of human walking. In practice, a complete gait cycle is defined as the interval between two consecutive heel strikes of the same foot, and it is commonly divided into two main phases: the stance phase and the swing phase.

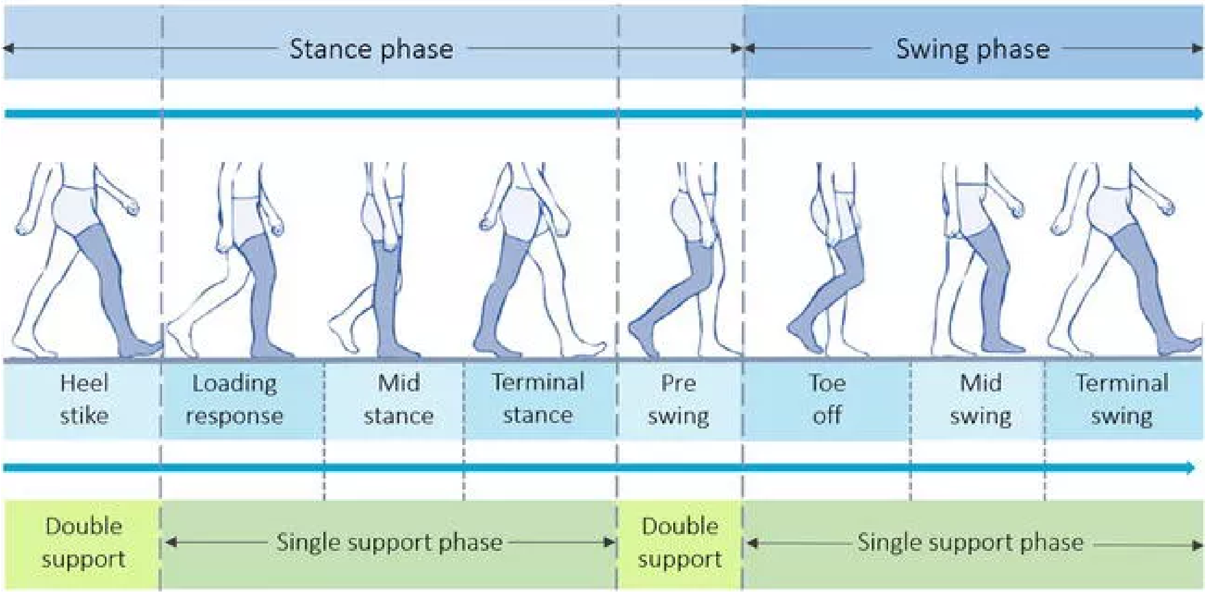

Figure 6.16: Subdivisions of the stance and swing phases of the gait cycle

The stance phase corresponds to the period during which the studied foot remains in contact with the ground. It includes both the single-support period, when the opposite foot is in swing, and the double-support period, when both feet support the body. This phase typically represents around 60% of the gait cycle [23][24].

The swing phase corresponds to the period during which the studied foot is moving forward. It accounts for approximately 40% of the gait cycle [23][24]. The swing phase begins with toe-off, continues through mid-swing, and ends at terminal swing, when heel strike occurs and the foot contacts the ground again.
Accurate stride length estimation relies primarily on the swing phase. In the collected sensor data, it is therefore essential to identify this phase in real time.

To identify the swing phase, we used the gyroscope data. Indeed, unlike other similar projects that placed the IMU on the foot (inside the shoe or on the bridge), our device is positioned on the ankle. As a result, its orientation varies in a characteristic way at each stage of the gait cycle, making the phases easier to recognize [13].

The following figure shows the similarity between the evolution of the inclination angle (unitless) and the different stages of the gait cycle of the corresponding foot. During the stance phase, the angle decreases and passes through 0°, which corresponds to the mid-stance (orange + yellow). Then, during the swing phase, it increases rapidly to reach the initial orientation (green). The decrease at the beginning of this phase corresponds to the toe-off stage, during which the foot must move backward to lift off from the ground.

To facilitate phase identification, the angle was measured with brief pauses between stages, corresponding to the constant, uncolored sections. In normal walking, these pauses would not appear, and the transitions between colored regions would be continuous.

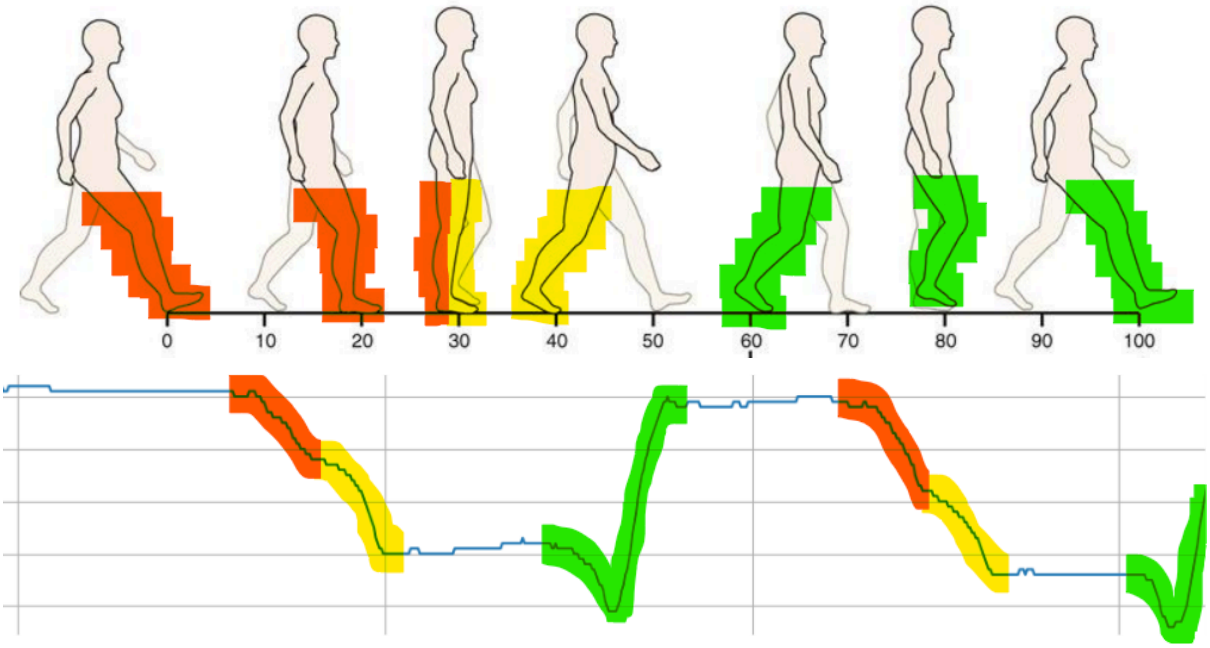

Figure 6.17 : Foot positions during the gait cycle with corresponding ankle inclination angle evolution

Thus, the swing phase can now be easily identified, allowing data to be processed specifically for this phase.

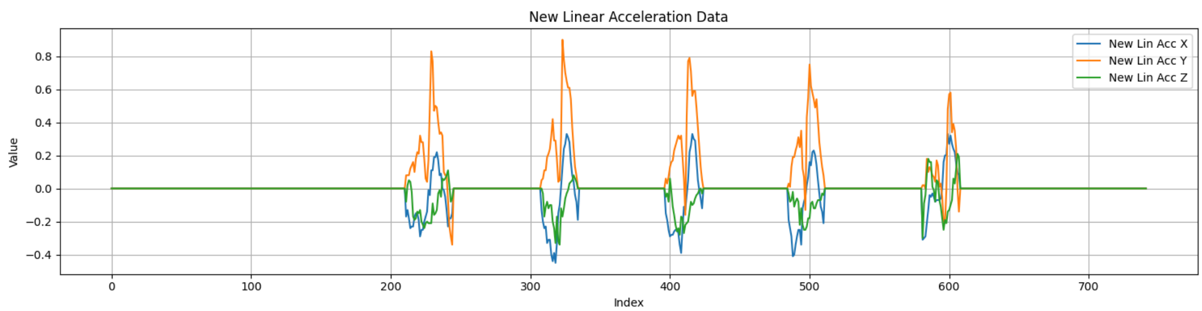

Figure 6.18 : Three-axis linear acceleration isolated to swing phase (zeroed during stance)

6.3.3. Position estimations and stride length calculations

Once the data have been separated, it becomes easier to integrate the accelerations into velocity and then into position. This segmentation allows computations to be performed over shorter time intervals, which significantly reduces error accumulation and mitigates issues associated with double integration.

Compared to simple Euler integration, we used trapezoidal integration :

Euler: **vn+1 = vn + an * Δt:**
Trapezoidal: **vn+1 = vn + (an+1 + an)/2 * Δt**

This method updates the velocity based on the average acceleration over a small time interval, providing a more accurate estimate than the simple Euler method while remaining sufficiently straightforward for implementation on an embedded system.

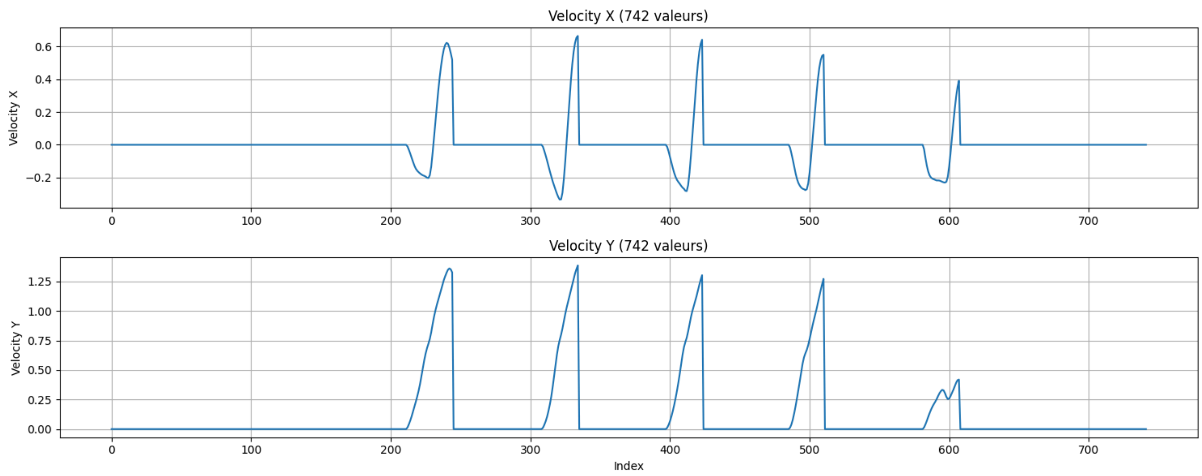

Figure 6.19 : Estimated velocity components along the x-axis (vx) and y-axis (vy)

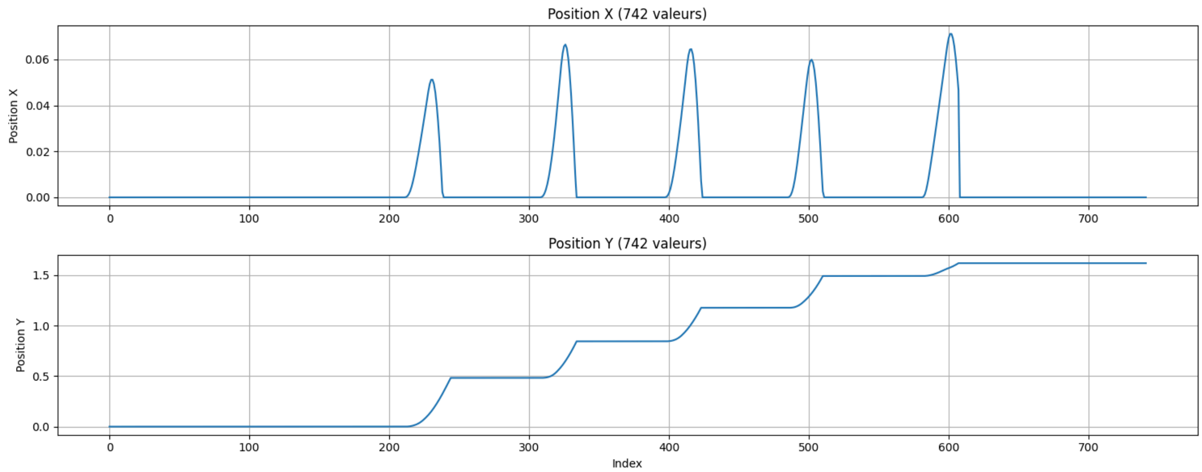

Figure 6.20 : Estimated position components along the x-axis (px) and y-axis (py)

To obtain an accurate position estimation, we adjusted the reference frame so that the x-axis points upward instead of toward the ground, while the y-axis represents the walking direction. The resulting trajectory shows a stair-step pattern, where the stride length corresponds to the magnitude of displacement at each step.

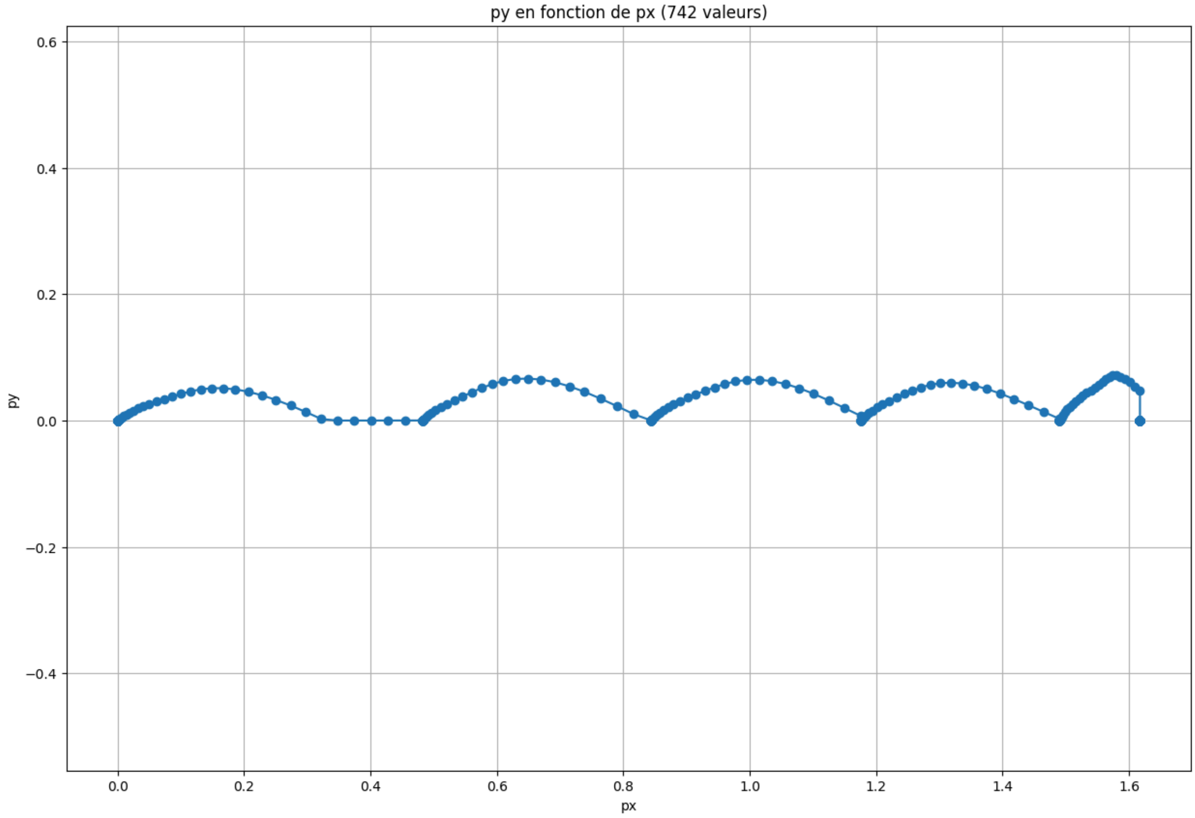

Figure 6.21 : 2D walking trajectory

As we are calculating the stride length asymmetry between the left and right legs, no scaling was applied to conserve processor power. The following diagram shows the results obtained in a realistic scenario where the subject walked for five minutes.

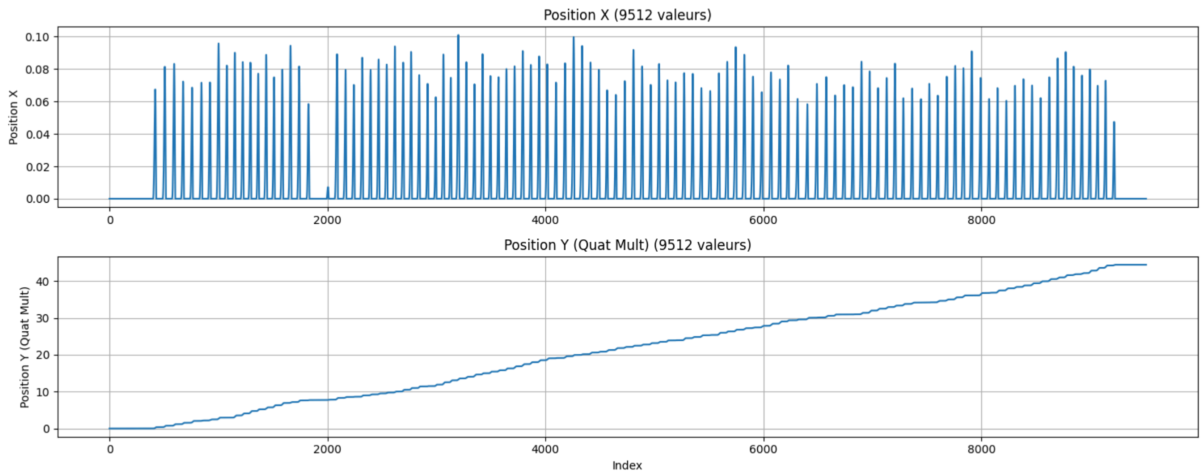

Figure 6.22 : Results from the 5-minute walking test

From this data we can calculate the stride length and send them via BLE to UI.

6.4. Wireless communication

6.4.1. Importance of wireless communication

Wireless communication is one of the main aspect of our TREK 2.0 bracelet, as it enables data transmission from the embedded sensors to an external device without any cables. This is crucial for our wearable gait analysis device: the absence of cables enhances comfort, freedom of movement, and safety for the user.

In practice, data from the IMU are transmitted after processing in real time to our web interface (on the computer or the phone). This allows instant analysis, live visualisation, and facilitates clinical or experimental monitoring.

Wireless communication therefore plays a key role in ensuring portability, ease of use, and responsiveness, while contributing to the compact and lightweight design of our TREK 2.0 system.

6.4.2. Choice of Bluetooth Low Energy (BLE) over Wi-Fi

We had two wireless options during this project: Wi-Fi and Bluetooth Low Energy (BLE). While Wi-Fi provides high bandwidth (typically 50–150 Mbit/s) and long range (up to 50–100 m indoors), it also demands high power consumption, often 150–250 mA during transmission, and requires a network infrastructure (router or access point), along with complex configuration steps (IP addressing, authentication, credentials, etc.). These factors are incompatible with the TREK 2.0 design goals of simplicity and autonomy.

BLE, on the other hand, was designed for low-power and short-range communication in wearables and connected devices. Its main advantages include:

- Very low power consumption: typical values range from 5–15 mA during transmission and only 1–10 µA in deep sleep, allowing extremely long battery life.
- Direct device-to-device connection, without requiring any Wi-Fi infrastructure. BLE pairing and data exchange occur directly via GATT services.
- Low latency, suitable for real-time sensor data: wake-up times around 6 ms, and effective communication latency often below 20–30 ms.
- Moderate data rate, fully sufficient for sensor information: BLE V4.x supports up to 1 Mbit/s, and BLE V5.x up to 2 Mbit/s.
- Short-range but stable communication, typically 10–30 m, and up to 100 m in BLE 5 Long Range modes.
- Broad compatibility with modern systems on native support such as Microsoft, Android or IOS

So we choose BLE because it offers the most appropriate solution: a lightweight, reliable, and energy-efficient communication method perfectly aligned with our bracelet's requirement

6.4.3. How Bluetooth Works in the System

Bluetooth communication in TREK 2.0 follows the Client/Server model defined by the BLE protocol.

On the bracelet (BLE server)
The embedded board acts as a BLE server. It advertises a service, identified by a unique UUID, containing several characteristics that hold sensor data such as accelerometer or gyrometer that we will process to obtain the positions... The server automatically sends updates to connected clients via BLE notifications.

On the computer/phone (BLE client)
The computer or web browser acts as a BLE client. Unlike headphones or keyboards, the TREK 2.0 bracelet does not appear in the operating system's Bluetooth list, because those menus only detect classic Bluetooth devices (using SPP, HID, or A2DP profiles).

So BLE has to use a distinct GATT (Generic Attribute Profile) structure, designed for structured and event-driven data exchange.

As a result, the connection cannot be established through standard Bluetooth pairing. Instead, it must go through software capable of communicating over GATT. That is why, in this project, our web app uses the Web Bluetooth API.

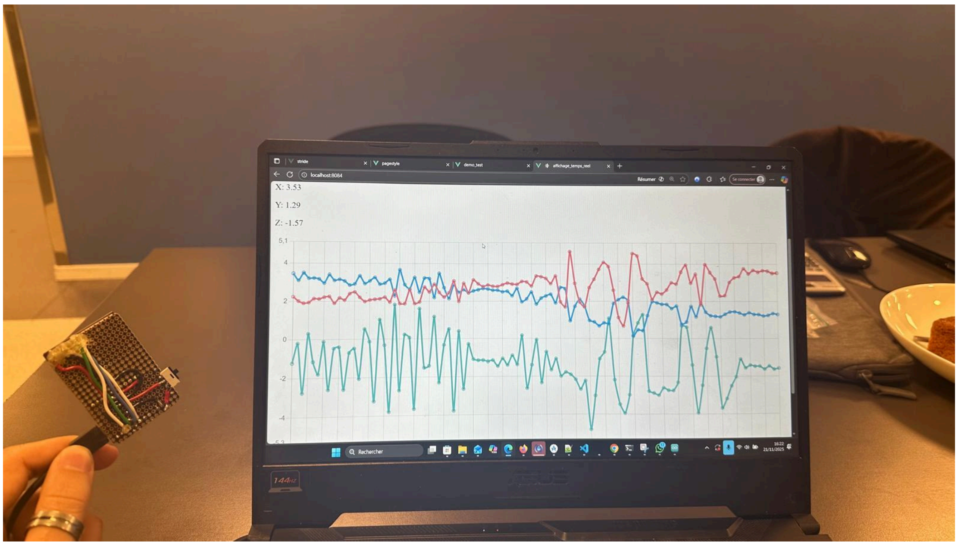

Figure 6.23 : Independence between the card(BLE client) and the computer (BLE server)

UUIDs and Security Context
Each BLE service and characteristic is identified by a UUID, allowing the client to locate and access specific data. During connection, the browser queries these UUIDs to determine available sensor characteristics. For security reasons, browsers only allow BLE access from secure HTTPS contexts or localhost. This prevents untrusted websites from scanning or interacting with nearby Bluetooth devices. Once connected, data is received as binary packets, decoded and displayed or processed in real time. For more details, these resources can be useful : [25] and [26].

6.5. User Interface design and application development

The user interface (UI) plays a crucial part in the overall user experience, as it is the first thing a user interacts with. During this project, attention was given to designing an interface that is both intuitive and user-friendly.

Choosing code :
For this project, we had many options to choose from, but as a team with limited coding experience, we decided to use Nuxt 4 as the foundation for our website/web app. Combining Nuxt 4 with HTML and JavaScript was challenging at first, but it proved to be well worth the effort. Nuxt provides a clean structure, automatic routing, and useful tools that make the development process easier and faster. Since Nuxt is built on Vue, our team can write template-style code that feels like enhanced HTML while still benefiting from reactive JavaScript and modern web features.

By taking these features into account, I decided to develop a website first rather than an app. This allowed us to clearly demonstrate the core functionality and the intended use of the product. After this, we were hoping to configure the code from the website into an app. This turned out to be a very difficult task. Unfortunately, it was something that we as a project group weren’t able to do with our skillset.

Since this was a completely new experience for the entire team, making a website offered greater stability and provided more opportunities to work on the project.

Implementation

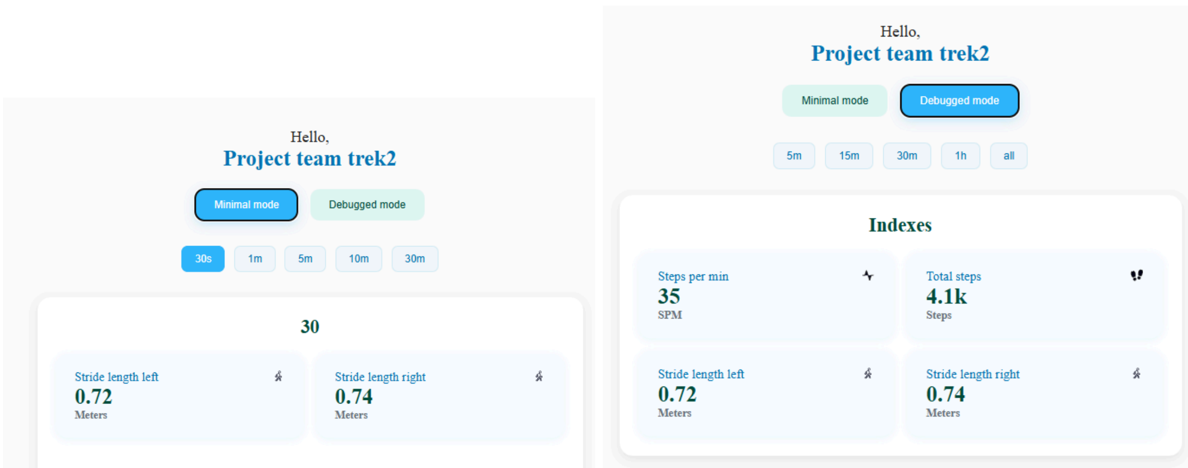

Figure 6.24: UI design

In the implementation (as shown in Visual Studio Code), we integrated functionality that allows the user to switch between Minimal Mode and Debugged Mode. Each mode provides access to five different time frames, enabling users to observe patient activity over different intervals. These time frames display valuable insights such as the number of steps taken and any deviations in movement patterns.

Real-Time Visualization
A local web application was developed to display real-time data, providing an overview of the patient's performance. The User Interface offers both the minimal and debugged modes, allowing users to toggle between simplified and extensive views as needed (see figures above).

Minimal Mode
The Minimal Mode focuses on essential metrics, presenting only the most important information for quick assessment. Displayed indexes include:
- Stride length of the left and right legs.
- A parabolic graph illustrating the deviation between the left and right leg.

This mode enables users to instantly identify differences or imbalances between the patient's left and right strides, ensuring fast interpretation without unnecessary data clutter.

Debugged Mode
The Debugged Mode provides an expanded view of patient performance, offering a deeper insight into movement dynamics. Displayed indexes include:
- Steps per minute
- Total steps
- Step deviation (left and right)

These indicators are numerical indicators and are being shown in two graphs:
1. A deviation graph, visualising the differences between the left and right leg.
2. A cumulative steps graph, showing the total number of steps taken by the patient over time.

This mode supports in-depth analysis, enabling healthcare professionals to monitor progress and identify potential gait irregularities more effectively.

6.6. Integration and system validation

The integration involves assembling the case with the Nicla Sense and the battery. The battery is secured in the bottom part of the case, and the PCB Board with all its components is positioned in the designated location on top. The upper part of the case is then screwed onto the bottom part, holding all components firmly in place. Afterwards, it is verified that all connectors are correctly positioned and the strap is attached, resulting in a fully functional overall system.

The system validation then checks whether the assembled prototype meets the planned requirements. This includes ensuring that the case fits properly on the foot, the battery supplies power correctly, the control elements such as the Bluetooth button and the On/Off switch function properly, and that the measurements from the sensors operate as intended.

Figure 6.25 : Integration of all the components in the case

7. Conclusion

7.1. Project conclusions and key findings

The development of Trek 2.0 has successfully transformed the previous prototype into a more capable, compact and user-friendly gait monitoring system. Whilst building on the foundation left by the previous group, our team implemented significant improvements across hardware, firmware, signal processing, wireless communication and user interface design.

The custom PCB and housing allowed us to reduce the total device size. Improving the power efficiency and reliability of the product ensured that we have created a gait monitoring system that supports recovery and daily function.

The development of a local website provided a medium for visualisation, which shows the time window analysis and mode selection.

With these improvements, Trek 2.0 is now presented as a complete and adaptable solution for gait monitoring, capable of processing patient information in real time according to user needs.

In the future, this bracelet could serve as an effective tool for tracking patient progress, allowing healthcare professionals to easily interpret the data displayed on the user interface and tailor care accordingly.

7.2. Skills acquired

Chethan :
During the EPS project, I applied my existing skills in PCB design and troubleshooting, prototype testing, and resource planning, ensuring the project stayed on track despite hardware issues. I also used my experience in team coordination, separating hardware and programming tasks to improve efficiency.

Jeya :
I learned a lot during this project, both in terms of management and technical skills. I worked on signal processing at a higher level than in my previous projects, facing significant hardware constraints, which allowed me to be fully immersed in the work of an embedded systems engineer. It also helped me understand and apply new algorithms, such as the Kalman filter, and see firsthand how they contribute in a real system.

Thomas :
During my engineering studies, I primarily focused on mechanical and electrical fields. However, during this EPS project, I was able to develop skills in a field that didn’t initially attract me: computer science. Starting with work on wireless communication, I learned more about Bluetooth communication, a technology that surrounds our daily lives. Additionally, I enhanced my coding skills by developing the user interface for the website we aimed to

create. I discovered a real passion for these tasks and found genuine satisfaction in accomplishing them.

Michael :
During the project, I further enhanced my skills in CAD design and 3D printing, especially through the practical creation of a functional prototype. I was responsible for the mechanical design, through which I learned how to coordinate decisions within a team. Taking responsibility for my own area, from design to the manufacturing of the prototype in an international team, was another key learning experience. Additionally, I was able to improve my English skills.

Luke :
During this project, I developed new skills that will support my future career as a mechanical engineer. I began with basic coding, which eventually grew into building a fully functioning website. This is something I never would have expected at the start of the project.
In addition to technical skills, I also improved my ability to work effectively in a team. Even when things didn't go my way, I stayed communicative and helped divide the workload to benefit the project. By applying the methods taught in class, our group gradually became a more efficient and cooperative team.

7.3. Reflections and overall experience

Chethan :
While the PCBs did work in the end and matched our use case, a smaller, thinner application would have better met the project's needs. In an ideal world, the programmer for the Arduino would be separate, and only the Arduino's CPU, RGB LED, and necessary passive components would be on the board.

The PCB itself, like any V1, can be improved. In a subsequent version, the following changes should be made.
- Flip the JST connector 180 degrees
- FLIP the MicroSD card reader 180 degrees or remove capacitors / resistors, as they are in the way of inserting an MicroSD card
- Remove R5, this resistor ties the TS to the OUT pin on the LDO for the charging circuit, causing the charger not to function correctly
- issue with button, needs to be rotated 90 degrees, or GND should be on the opposite pin

These errors resulted in a few wasted boards and Nicla's being unusable (the pins on the Arduino Nicla board were soldered to what was thought as a completed board, as while the initial tests were passed, further errors were found upon a second validation). We would also recommend a future group attach pin headers (assuming they do not make a custom Arduino) only to the PCB and not the Nicla during testing.

In addition, we would recommend designing and purchasing boards early on in the project, although we recognize this is dictated by other project dependencies. Assuming purchasing custom boards is not feasible, we would recommend purchasing at least additional programming boards and PCBs. Again, both in advance if possible, as Arduinos can be purchased at the very beginning of a project, and more duplicates of PCBs give the project additional slack in case something goes wrong. Purchasing additional PCBs allowed for the project's success despite testing and validation errors (ideally, these errors don't happen at all, but the additional PCBS kept our group from having an unusable prototype).

Jeya :
C'était une nouvelle expérience, la première fois que je travaillais avec des personnes que je ne connaissais pas, et encore plus avec des nationalités différentes. il y a eu des hauts et des bas, mais j'ai été content de voir une équipe soudée. j'ai aussi beaucoup appris sur les autres, leurs cultures, leurs façons de travailler et leurs connaissances.

Thomas :
During this project, I was able to discover new ways of working by interacting daily with people from different cultures. At first, it was quite unsettling because I had been taught to work in a certain way throughout my schooling. But by embracing these new working methods, I was able to open myself up more to the world around me, improving both my listening skills and my ability to learn.

Michael :
The work on the project was characterized by close collaboration within the team. Since the housing had to be precisely matched to the internal components, continuous coordination was necessary with the team members responsible for electronics and sensor technology. Many design decisions could therefore only be made once certain technical details had been determined. A challenge was that we, as exchange students, were not always available at the same time. Due to absences, tasks sometimes had to be restructured or rescheduled to avoid further delays in the project. Communication of technical terms was challenging, as it was conducted in English, which was a foreign language for most team members. This allowed us, as a team, to make continuous progress and move closer to our goal of creating a functional prototype.

Luke :
I entered the project with a background in mechanical engineering, a field heavily focused on calculations, design, and understanding how a product behaves or appears.However, my primary responsibility in this project was working on the user interface, an area completely new to me. This made the project both challenging and exciting.I spent a significant amount of time researching where to begin and eventually started working with Nuxt 4 and Vue to create a smooth and pleasant user experience.

Despite the excitement. I faced personal challenges. My health occasionally forced me to step back, and I made sure to communicate openly with the group whenever this happened. Near the end of the project, I was unfortunately involved in a car accident, which required me to step away for several weeks. This was difficult, but I am very grateful that Thomas stepped in and supported the team during that time.Spending long hours behind a screen became particularly tough, which may be reflected in my (latest) work.

In the end, everything came together. In my honest and professional opinion, I highly recommend participating in a project like this with a diverse group of international students, just as we did. It was a valuable and rewarding experience.

8. References

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9. Appendix

The AI language model ChatGPT was used as a supportive tool during the preparation of this report. Its assistance was limited to improving phrasing, enhancing clarity, and correcting grammar and language usage. All technical content, analysis, and conclusions presented in this report were generated by the authors, with ChatGPT serving only as a writing aid and not influencing the scientific or technical work.

Power Calculations

Assumptions
Operation: 16 hr active + 8 hr sleep
Battery: Lithium Ion batter, 3.7 V

Active
IMU (active): 0.50 mA
BLE (nRF52 radio): peaks ≈5.3 mA TX / 5.4 mA RX;
  on-air 3 ms per 100 ms interval (≈1.5 ms TX + 1.5 ms RX)
  BLE Avg = (5.3 * 1.5 + 5.4 * 1.5) / = 0.1605 mA
microSD (batched every 1 s during active):
  Peak Current; 100 mA   Idle Current: 0.2 mA
    per-record write time; 3 ms   overhead per burst; 10 ms   10 records/burst
  Burst Time = 10(3 + 10 = 40ms, or 0.04 s
    AVG Current = 0.2 + (100-0.2) *0.04 = 4.192 mA
  Duty during active = (40/1000=0.04)

Sleep
  Nicla standby 0.46 mA, SD idle 0.20 mA, Net 0.66 mA

Per-day mAh
  IMU: 0.5 mA * 16 hr = 8 mAh
  BLE: 0.1605 mA * 16 hr = 2.68 mAh
  SD: 4.192 mA * 16 hr = 67.072 mAh
  Sleep: 0.66 mA * 8 hr = 5.28 mAh
  Total: (8 + 2.68 + 67.072 + 5.28) mAh = **82.82 mAh**
.Total mWh = Battery voltage * Total mAh = 3.7v * 82.92 = **306.8 mWH**

The vast majority of the power consumption is from the SD card at roughly **81%** of the power cost. This is non-negotiable for our client, but alternative solutions such as cloud storage or increased processor cache size to decrease microSD access rates are recommended.

MicroSD Card Memory Calculation

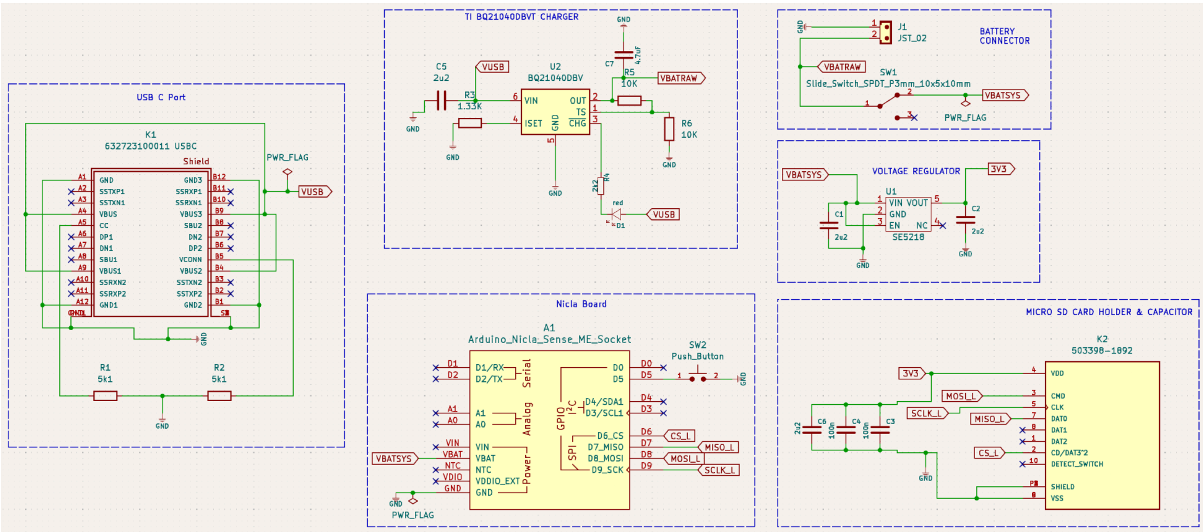
Is the SD card's memory capacity a limiter?   No.

The SD card we have is 32 GB. Assuming we send 5 floats (3 for sensor data, 1 for timestamp, and 1 for ID) every 100ms (for our 10 Hz Sampling rate), we get:

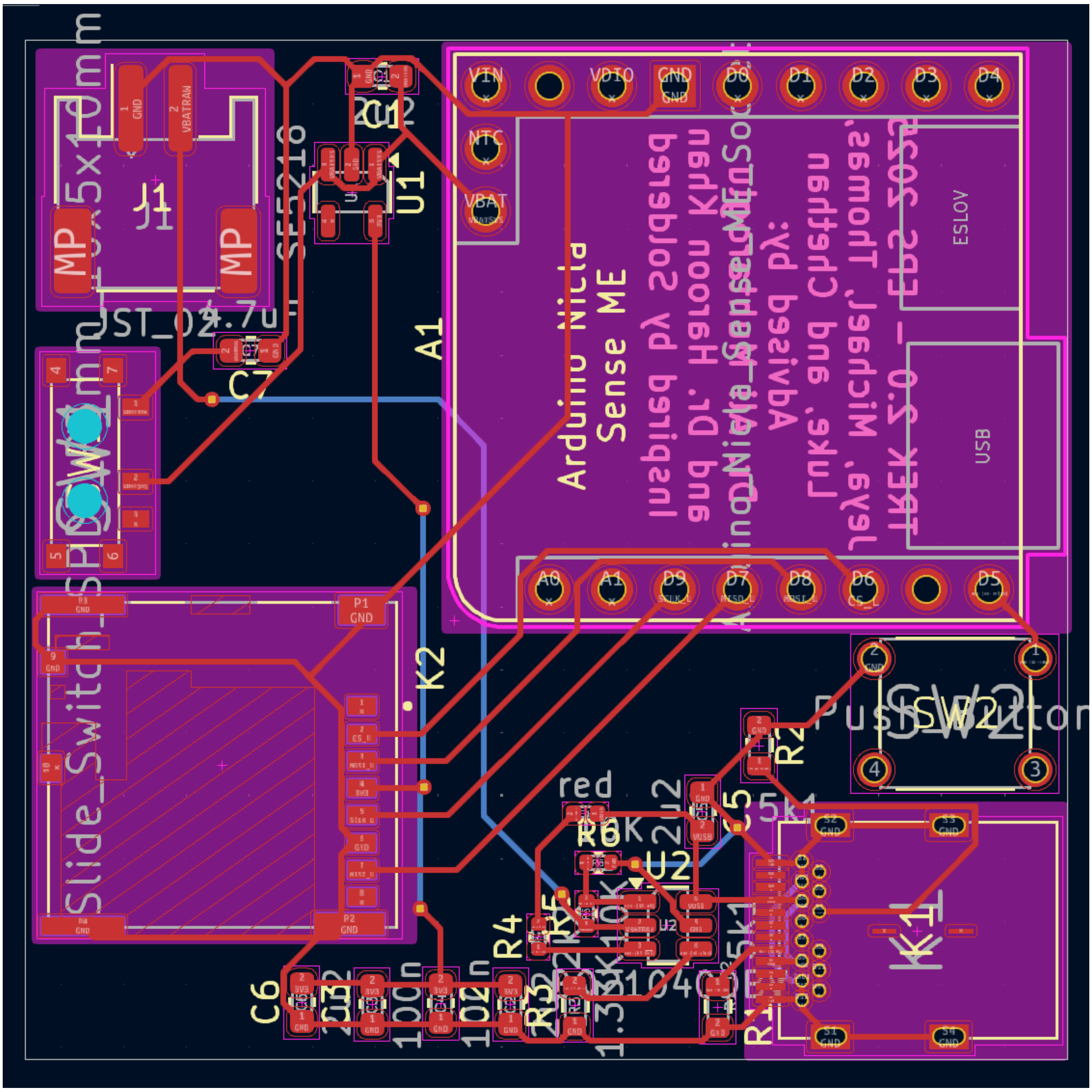
5 floats per sample = 20 Bytes / sample
10 samples per second = 200 Bytes per second
200 Bytes per second 60 60 24 = 17.28 MB / Day

The SD Card has 32,000 Mb. 32000 /17.28 = 18,151 days or **more than 5 years of storage**

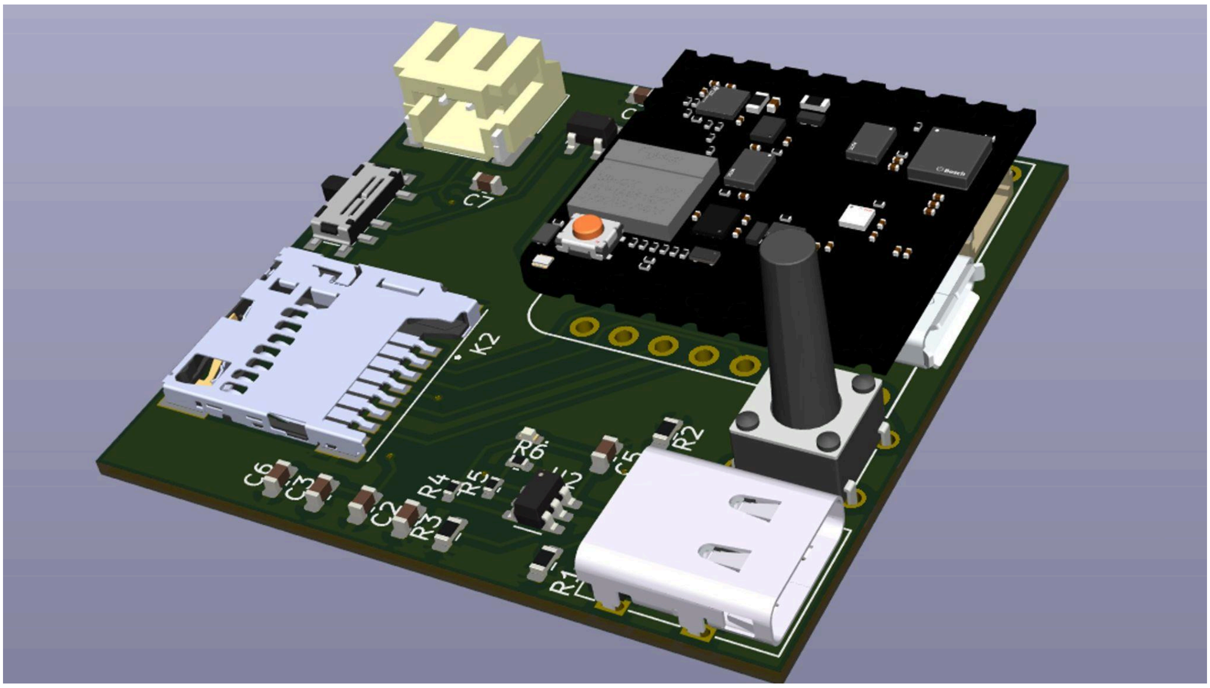
PCB Images



Printed Circuit Board Schematic, Kicad



Printed Circuit Board Layout, Kicad



Printed Circuit Board 3D Model, Kicad